



**Maastricht University**

**Master's Internship Report**

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**From Indicators to Typologies : Unsupervised Machine Learning for Climate Risk Clustering in Europe**

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*Realized by :*  
**Catella Investment Management**



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# Abstract

This project presents a comprehensive, data-driven framework for European climate clustering at the NUTS3 regional scale, designed to challenge and evaluate existing expert-defined climate typologies. Using a transparent and modular pipeline built around *Hazard–Exposure–Vulnerability* (HEV) indicators, we reconstruct the core components of regional climate risk and use them as input to an unsupervised machine-learning system for regional classification.

Despite not having access to the full proprietary dataset used in the RESIN European Climate Risk Typology, our experiments demonstrate that algorithmic clustering can reproduce its broad structure and spatial organisation using open and reproducible indicators. We systematically evaluate multiple embedding–algorithm configurations (PCA, KPCA, UMAP combined with KMeans, Agglomerative Clustering, Spectral Nearest-Neighbour clustering, and Gaussian Mixture Models). Model performance is assessed using internal validation metrics (Silhouette, Calinski–Harabasz, Davies–Bouldin) and external agreement with the RESIN expert labels (ARI, NMI, V-measure), complemented by a cross-model consensus analysis to identify stable regional groupings.

Across the best-performing models, we obtain approximately 20% exact label agreement with the expert typology and, more importantly, recover highly similar large-scale spatial patterns across Europe. The resulting clusters align with climatic gradients, dominant hazard regimes, and socio-environmental conditions, indicating that unsupervised algorithms can uncover the same latent regional structures that are typically derived through expert judgement, provided that the feature set captures the HEV dimensions in sufficient breadth and quality.

This work demonstrates that:

- Expert climate typologies are reproducible with unsupervised learning using open-data-driven Hazard, Exposure, and Vulnerability indicators.
- Clustering can function as an independent, quantitative validation layer for expert assessments by highlighting both agreement and systematic divergence.
- The modular pipeline generalises to other regional typology problems, as shown by:
  - a European flood-scenario clustering analysis that produced coherent regional patterns under perturbed hazard assumptions, and
  - a German housing affordability study that yielded stable and interpretable socio-economic typologies.

Overall, the results highlight the reliability of unsupervised clustering for regional classification when paired with transparent feature engineering and rigorous validation. The proposed

methodology provides a scalable foundation for future climate-risk assessment, scenario analysis, and decision support, enabling researchers, asset managers, and policymakers to replicate, challenge, or extend expert-derived models using reproducible and data-driven tools.

# Chapter 1

## Introduction

We are now approaching the end of the first quarter of the 21st century, and the signs of climate change are no longer distant predictions; they are already happening. One clear signal is sea-level rise: since the start of the satellite record in 1993, global mean sea level has increased by roughly 10 cm, and the rate of rise has more than doubled over the same period.<sup>1</sup> For real estate asset managers, this is not just a scientific statistic; it is a financial warning.

Real estate portfolios are long-term by nature. A single severe climate event, whether it happens once a decade or once in a generation, can significantly damage asset value, disrupt operations, and reduce long-term returns. The 2021 flood disaster in western Germany illustrates this clearly: the event caused tens of billions of euros in total damage, while insured losses were only a fraction of that total.<sup>2</sup> In practice, this creates a serious investment problem: when coverage is incomplete (or absent), a large share of losses remains with owners, landlords, and ultimately investors.<sup>3</sup> It also exposes a deeper issue: risk assessments calibrated on historical patterns and “stable” return periods can fail under accelerating climate change, where hazard intensity, timing, and spatial footprints shift faster than traditional models and market assumptions.

Because of this, investors and asset managers require a structured, evidence-based approach to understanding how climate hazards will impact their portfolios both today and in the decades to come. To address this, we needed a comprehensive study: one that not only examines future climate forecasts but also assesses the exposure of each region and the vulnerability of the community and infrastructure surrounding the assets.

A central reference point for our work is the RESIN project by the European Commission. RESIN created a robust framework to classify European regions into climate-risk typologies using expert-driven logic. Their work has provided a strong conceptual basis for understanding regional hazard, exposure, and vulnerability patterns.

At Catella Investment Management, our goal was to build on this foundation and take it to the next level.

We asked a simple but essential question:

Can this typology approach, initially built by experts, be recreated algorithmically using only data?

To answer this, we experimented with a wide range of clustering techniques, including both linear and non-linear approaches. We tested hundreds of model combinations, evaluating

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1. <https://sealevel.nasa.gov/news/282/nasa-analysis-shows-unexpected-amount-of-sea-level-rise-in-2024/>

2. <https://nhess.copernicus.org/articles/23/525/2023/>

3. <https://www.cleanenergywire.org/news/many-germans-still-underinsured-against-floods-insurance-industry-warns>

internal clustering quality and comparing results to the expert labels produced by RESIN.

The findings have been surprisingly strong.

Even without the complete dataset used in the original RESIN study, we were able to replicate similar spatial patterns across Europe. About 20 percent of the expert labels matched directly, and the overall clustering behavior closely followed the same regional structures identified by domain experts.

This confirms two key points:

Climate-risk typologies can be reproduced algorithmically, even with limited or alternative data sources.

Asset managers can integrate climate-risk clustering into investment analysis, making it a quantitative and reproducible part of decision-making.

The work in this report demonstrates how data-driven methods can support and enhance climate risk assessment. More importantly, it lays the groundwork for a stronger, more forward-looking investment strategy, one where climate risk is not treated as an optional add-on, but as a primary factor in understanding the long-term performance of real-estate assets across Europe.

## Chapter 2

# Problem Statement

Climate change is reshaping the physical risk profile of European residential real estate. Hazards such as river and surface flooding, coastal inundation, droughts, heatwaves, wildfires, and sea-level rise are increasing in frequency and severity, while the financial consequences are amplified by long holding periods and imperfect risk transfer mechanisms (e.g., underinsurance and coverage exclusions in high-risk areas). For an investment manager like Catella, this presents a concrete decision-making challenge: climate risk must be measured and compared consistently across hundreds of European regions, using a methodology that is transparent, reproducible, and adaptable to new data and scenarios.

The core technical problem is that relevant climate-risk information is fragmented across heterogeneous sources and formats. Datasets differ in spatial resolution (grid vs. administrative boundaries), temporal scope (historical vs. projections), and definitions of hazards and impacts. Additionally, there is no universally accepted "ground truth" dataset that categorizes European regions by climate risk class for investment purposes. Existing expert-defined typologies, such as the European Commission's RESIN framework, provide valuable reference classes, but they are not designed as an operational, continuously updatable system for large-scale portfolio screening and monitoring.

This creates a practical gap for investment teams and risk analysts:

- consolidating multi-source hazard, exposure, and vulnerability indicators into a unified regional dataset,
- translating these indicators into quantitative and comparable risk measures suitable for portfolio analysis,
- evaluating whether unsupervised clustering can reproduce, validate, or systematically challenge expert-defined typologies, and
- presenting results in an interpretable form that supports both top-down strategy (regional allocation and risk concentration) and bottom-up decisions (acquisition, disposition, and asset-level due diligence).

In summary, the problem addressed in this report is the development of a reliable, data-driven, and explainable method to classify European NUTS3 regions by physical climate risk using Hazard–Exposure–Vulnerability (HEV) information, such that climate risk can be integrated directly into Catella's investment and portfolio decision processes. This includes building a large-scale climate-risk dataset, designing a composite risk scoring layer, benchmarking multiple clustering approaches against expert typologies, and delivering the outputs through an interactive dashboard for decision support.

## Chapter 3

# Research Questions

### 3.1 Questions Outline

Climate change is already affecting the risk profile of European regions, but for investors and asset managers it is not straightforward to translate heterogeneous climate data into actionable insight. This study therefore investigates a sequence of research questions that move from (i) data integration, to (ii) risk scoring, to (iii) typology construction, and finally to (iv) interpretability and (v) operational decision support.

#### RQ1: Data Integration and Representation

**RQ1:** *How can large, heterogeneous climate datasets (Hazard, Exposure, Vulnerability) be consolidated into a unified, machine-learning-ready dataset at NUTS3 resolution?*

Climate information is distributed across multiple sources and differs in format, spatial resolution (grids/rasters vs. administrative regions), units, and time horizons. Indicators describe physical hazards (e.g. flood depth, heat, drought), exposure (e.g. land cover, population, built-up area), and vulnerability (e.g. socio-economic and infrastructure proxies). This question focuses on establishing a consistent data foundation:

- Aggregating gridded/raster climate data to NUTS3 regions in a statistically sound and interpretable way.
- Harmonising units, scales, and temporal definitions across providers.
- Handling missing values, outliers, and strongly skewed distributions to avoid bias in downstream models.
- Structuring the final feature set so that it supports both composite scoring and unsupervised learning.

#### RQ2: Climate Risk Scoring

**RQ2:** *How can a transparent composite physical climate-risk score be designed for regions, balancing expert judgement with data-driven evidence?*

Asset management workflows often require a comparable summary measure, while climate risk remains multi-dimensional. This question examines how to combine hazard, exposure, and vulnerability indicators into an interpretable score per region:

- Grouping indicators into meaningful components (Hazard vs. Exposure vs. Vulnerability) and applying appropriate normalisation.
- Defining an expert-informed weighting scheme reflecting the relevance of hazards for residential real estate.

- Testing and refining weighting choices using statistical and machine-learning evidence (e.g. variance-based weights, PCA structure, model-driven importance).

The objective is a score that is usable for investment workflows while remaining explainable in terms of its drivers.

### RQ3: Algorithmic Typology Construction

**RQ3:** *To what extent can dimensionality-reduction and clustering methods reproduce or challenge expert-defined climate typologies (e.g. RESIN) when applied to a broad HEV dataset?*

Expert typologies provide a valuable benchmark but are based on fixed indicator sets and expert logic. This question tests whether unsupervised learning can recover comparable structures and potentially reveal additional patterns:

- Identifying suitable representations of the climate feature space (e.g. linear PCA vs. non-linear embeddings).
- Comparing clustering algorithms (e.g. KMeans, Agglomerative, Spectral NN, GMM) using internal validation metrics.
- Benchmarking agreement against RESIN or other expert labels using external metrics (ARI, NMI, V-measure).
- Analysing whether algorithmic clusters reveal stable groupings not explicit in expert typologies, and how these can be interpreted.

### RQ4: Interpretability and Scenario Extensions

**RQ4:** *How can the resulting typology be made interpretable for non-technical stakeholders, and how transferable is the pipeline to additional scenario studies?*

For practical use, clusters must be explainable and reusable beyond the initial experiment. This question focuses on:

- Explaining cluster meaning via profiles and indicator contrasts (what defines each group in HEV terms).
- Linking cluster assignments back to original indicators using interpretable summaries (e.g. feature importance, cluster centroids/prototypes).
- Extending the same modular pipeline to additional scenarios (e.g. a dedicated European flood study and a German housing affordability clustering) without redesigning the full method.

### RQ5: Business Integration and Dashboard Design

**RQ5:** *How can risk scores and typologies be integrated into an interactive dashboard to support acquisitions, dispositions, and strategic portfolio planning?*

Analytical results create value only if they are accessible and usable for decision-making. This question addresses operational integration:

- Designing multi-level views (European overview, country context, and asset-linked regional insights).
- Linking regional risk information to specific assets to enable both top-down screening and bottom-up due diligence.
- Allowing exploration of assumptions (e.g. weight choices, scenario switches) and observing the impact on scores and typologies.

- Ensuring the interface and explanations are simple for daily use while retaining the ability to drill down for technical users.

Answering RQ5 connects the technical methodology to real investment workflows and demonstrates practical decision support.

# Chapter 4

## Literature Review

### Introduction

This chapter reviews the main strands of literature that underpin this thesis. The work sits at the intersection of: (i) European climate risk typologies such as RESIN, ESPON and EEA frameworks, (ii) hazard–exposure–vulnerability models from the IPCC, (iii) regional climate risk quantification and composite indicators, and (iv) clustering and validation methods used in climate science and socio-economic typologies.

The aim is not to provide a full survey of all related work, but to position the thesis within these key debates and to show how the proposed data-driven typology builds on, and differs from, existing approaches.

### 4.1 Climate Risk Typologies for European Regions and Cities

#### 4.1.1 RESIN and the European Climate Risk Typology

A central reference for this thesis is the European Climate Risk Typology (ECRT) developed within the RESIN project. The typology classifies European regions into groups with similar climate risk profiles, based on a combination of hazards, exposure and vulnerability indicators, and is designed as a practical tool for adaptation planning and resilience work in Europe [9, 18]. It provides both a conceptual framework and a set of maps at the NUTS3 level. European institutions and local authorities have already used it as a reference for regional climate risk.

RESIN is relevant for this thesis in three ways. First, it demonstrates that a typology approach is useful for communicating climate risk at regional scale. Second, it offers a set of expert-defined climate risk types that can be used as an external reference when developing new, data-driven typologies. Third, it shows how hazard, exposure and vulnerability can be combined in a practical way to support adaptation decisions.

#### 4.1.2 ESPON CLIMATE and EEA Frameworks

The ESPON CLIMATE project extends this type of work to a broader territorial perspective. It develops pan-European maps and typologies of climate change vulnerability, combining indicators of exposure, sensitivity and adaptive capacity at the regional scale [8]. ESPON distinguishes between different types of regions based on impacts on the economy, infrastructure and society, and their capacity to mitigate and adapt. The resulting typologies are used to support EU cohesion and regional policy.

The European Environment Agency (EEA) has also produced a series of vulnerability and impact maps for Europe, for example for river and coastal flooding, heatwaves and droughts. These products often apply the IPCC risk framework at continental scale, combining climate

hazards with information on exposure and adaptive capacity to derive composite vulnerability indices for European regions [6].

Together, RESIN, ESPON and EEA work show that:

- climate risk can be meaningfully summarised at regional level,
- typologies are a useful communication tool for policy and planning, and
- hazard, exposure and vulnerability should be considered jointly rather than in isolation.

However, most of these frameworks are based on expert judgement, fixed indicator sets and rule-based classification. They do not make systematic use of modern clustering techniques or explicit external validation metrics, which is one of the gaps addressed by this thesis.

## 4.2 Hazard–Exposure–Vulnerability Frameworks

### 4.2.1 IPCC AR5 / AR6 Risk Concept

The conceptual basis for this thesis comes from the IPCC risk framework. In the Fifth Assessment Report (AR5), Working Group II defines risk as a function of climate-related hazards, exposure and vulnerability, where vulnerability captures both sensitivity and adaptive capacity [7]. Hazards capture the physical aspects of climate change (e.g. extreme heat, heavy precipitation, floods), exposure describes the presence of people and assets in harmful locations, and vulnerability reflects the propensity to suffer harm given a hazard.

This framing is widely adopted in climate impact and adaptation research, and later reports such as AR6 refine it further but keep the same core idea of risk as a combination of these three components [14]. The EEA and ESPON frameworks mentioned above are direct applications of this concept at European regional scale.

### 4.2.2 Operationalising Hazard–Exposure–Vulnerability

A large body of work has focused on turning this conceptual risk framework into operational indicators and maps. For hazards, typical indicators include:

- climate indices derived from gridded climate model output (e.g. mean temperature change, summer days, heavy precipitation days),
- flood depth or inundation extent for different return periods, and
- drought indices and other measures of hydrological stress.

Exposure is usually captured by population, land use, built-up area or asset values in hazard-prone zones, while vulnerability is described using socio-economic indicators such as age structure, income, education, health status and access to services [7, 14].

These studies show that the hazard–exposure–vulnerability framework can be implemented in different ways depending on data availability and policy needs. This thesis follows the same logic, but uses it as the basis for a machine-learning pipeline rather than a purely rule-based classification.

## 4.3 Regional Climate Risk Quantification

### 4.3.1 Composite Indicators and Indices

Several European and global studies have developed composite indices to summarise climate impacts or vulnerability at regional scale. ESPON CLIMATE combines multiple indicators into a vulnerability index that reflects both potential impacts and adaptive capacity [8]. The EEA has produced indices for heatwave risk, flood risk and agricultural drought, often by normalising and weighting indicators and aggregating them into a single score [6].

Methodologically, these indices typically follow four steps:

1. selecting indicators for hazard, exposure and vulnerability,
2. normalising indicators to make them comparable,
3. applying Normalisation techniques, and
4. aggregating indicators into composite scores, for example, via weighted sums or simple averages.
5. Finally labelling the classes (often based on expert judgement)

The benefit of such indices is that they offer a simple summary measure for communication and comparison across regions. The downside is that the choice of indicators and weights is often subjective and rarely tested against alternative specifications.

### 4.3.2 From Indices to Data-Driven Typologies

More recent work links composite indicators to typologies or clusters. For example, the European Climate Risk Typology uses a structured set of hazard–exposure–vulnerability indicators to classify regions into a small number of risk types [9, 10]. Other studies have derived vulnerability or impact typologies by applying threshold rules to index values or by combining indices with expert knowledge.

This thesis builds on that idea but moves one step further towards data-driven methods. Instead of hand-coding rules or thresholds, it uses unsupervised learning to identify clusters directly in the indicator space, and then compares these clusters to an existing expert typology.

## 4.4 Clustering and Typologies in Climate and Socio-Economic Research

### 4.4.1 Clustering in Climate Science and Regionalisation

Clustering has been widely used in climate science to identify regions with similar climate characteristics or projected changes. For example, several studies have applied k-means clustering to climate model output to define homogeneous regions of temperature and precipitation change over Europe [2, 16]. Others cluster climate indices to identify typical patterns of extremes or combined hazard regimes.

More recent work has proposed clustering methods that better respect the spatial and temporal structure of climate data. Structural k-means (S k-means), for instance, incorporates information about spatio-temporal similarity and provides a dedicated framework for quantifying clustering uncertainty in climate applications [4]. There are also studies that cluster combined climate and air quality indicators (e.g. ground-level ozone and temperature) to identify regions that share similar exposure to compound hazards [11].

In most of these examples, clustering is applied either to climate variables alone or to a small set of indices. The focus is on regionalisation of climate change signals rather than full risk, which would also include socio-economic factors.

### 4.4.2 Clustering in Socio-Economic and Vulnerability Typologies

Beyond purely climatic variables, clustering is also used to build socio-economic typologies. ESPON and related projects have used cluster analysis to group regions with similar economic structures, demographic trends or levels of vulnerability to climate change [8]. In urban studies, clustering has been applied to income, housing, migration and labour-market indicators to classify cities into functional types.

These studies show that:

- unsupervised learning is a natural tool for discovering regional types when no ground-truth labels exist, and
- cluster-based typologies can be useful for policy, especially when combined with maps and narrative descriptions of each type.

However, many applications rely on relatively simple feature sets and do not always combine climate, exposure and vulnerability in a single pipeline. External validation against independent typologies is also rare.

#### 4.4.3 Methodological Choices: Dimensionality Reduction and Algorithms

From a methodological perspective, clustering often starts with dimensionality reduction. Principal Component Analysis (PCA) is widely used to summarise sets of correlated indicators into a smaller number of components while retaining most of the variance [12]. In climate applications, PCA is frequently applied to climate indices or model output before clustering, to avoid issues with high dimensionality and multicollinearity [2, 16].

On the clustering side, standard algorithms such as k-means, hierarchical clustering and Gaussian mixture models are still the workhorses [13, 21]. Their appeal lies in their simplicity, availability in standard software and relatively easy interpretation. More advanced methods, such as spectral clustering or density-based approaches, are sometimes used when cluster shapes are complex [17, 5], but they are less common in policy-oriented regional typologies because their results can be harder to explain.

This thesis follows a similar pattern: PCA is used to build a low-dimensional representation of the hazard–exposure–vulnerability space, and standard clustering algorithms are applied in that space. The focus is on robust preprocessing and careful validation rather than on developing new clustering algorithms.

### 4.5 Validation of Clustering and Regional Typologies

#### 4.5.1 Internal and External Validation Metrics

Cluster validation is an important and sometimes neglected topic in applied work. Internal validation metrics, such as the silhouette coefficient, Calinski–Harabasz index and Davies–Bouldin index, evaluate the compactness and separation of clusters based on distances in feature space [20, 1, 3]. They are useful for comparing candidate solutions (e.g. different numbers of clusters) without needing external labels.

External validation metrics compare a given clustering to a reference partition. Common measures include the Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI), which quantify agreement between two partitions while correcting for chance. These metrics have been used widely in machine learning and have also started to appear in climate and environmental clustering studies.

V-measure is another entropy-based external cluster evaluation measure that combines homogeneity and completeness in a single score [19]. While it is more common in text and language applications, the underlying logic is general and can be applied to any partition comparison.

#### 4.5.2 Consensus Clustering and Stability

Beyond static validation metrics, consensus clustering methods try to assess the stability of clusters under resampling. Monti et al. propose a resampling-based consensus clustering framework that aggregates the results of multiple clustering runs into a consensus matrix and uses this to assess the robustness of cluster assignments and the choice of the number of clusters

[15]. This approach has become popular in genomics and other fields where sample sizes are small and noise is high.

More recent work has started to investigate clustering uncertainty specifically for climate data, combining structural similarity measures with entropy-based uncertainty metrics [4]. These approaches highlight that, in complex high-dimensional problems, it is important not only to report a single clustering but also to understand how stable it is under data perturbations and modelling choices.

In this thesis, internal validation metrics and external comparison with the expert-based typology are used as the main tools to assess cluster quality. Full consensus clustering is beyond the scope of the current work, but the idea of stability and robustness is reflected in the sensitivity analyses on indicators, scaling choices and algorithms.

## 4.6 Positioning of This Thesis in the Literature

The literature reviewed above shows that:

- European climate risk typologies already exist, but they are mainly expert-driven and not tightly integrated with modern clustering and validation methods.
- The hazard–exposure–vulnerability framework from the IPCC provides a clear conceptual basis for risk assessment, but there is room to explore more systematic, data-driven ways of combining these components.
- Clustering is widely used in climate and socio-economic research, yet many applications focus on either physical climate variables or socio-economic indicators, not both together in a single risk-oriented feature space.
- Validation of regional typologies is often limited to qualitative judgment; explicit external metrics such as ARI, NMI or V-measure are still relatively rare in this context.

This thesis contributes to this literature by:

1. Building a large-scale hazard–exposure–vulnerability dataset at NUTS3 level for Europe using a transparent preprocessing pipeline.
2. Developing a purely data-driven climate risk typology using dimensionality reduction and clustering, and comparing it quantitatively to an established expert-based typology.
3. Using standard internal and external validation metrics to assess cluster quality and agreement with expert classifications.
4. Demonstrating that the same methodological backbone can be reused for other scenario studies, such as a dedicated flood analysis and a housing affordability clustering, which are documented in the appendix.

In this way, the thesis positions itself as a bridge between expert-based European climate risk typologies and modern, data-driven methods from machine learning, with a strong focus on interpretability and practical use in real-estate investment decisions.

# Chapter 5

## Methodology

### 5.1 Methodology

This chapter describes the end-to-end pipeline used to construct a regional climate-risk typology for European NUTS3 regions and to answer the research questions. The workflow follows a clear sequence: (i) select and prepare data sources, (ii) construct hazard, exposure and vulnerability indicators at NUTS3 level, (iii) build a robust preprocessing pipeline, (iv) learn low-dimensional representations, (v) run and compare clustering techniques, (vi) evaluate the resulting clusters, (vii) interpret and explain the clusters, and (viii) test sensitivity to key modelling choices.

In practice, most of the project effort went into steps (i)–(iii). The input datasets differ in formats, coordinate systems, temporal coverage and data quality. Making them consistent and reliable is therefore treated as a first-class part of the methodology rather than a minor preprocessing step. Figure 5.1 provides a high-level overview of the end-to-end methodology, from data integration and indicator engineering to clustering, interpretability, and business integration.

#### 5.1.1 Data Sources

The analysis is carried out at the NUTS3 regional level for Europe. All indicators, regardless of their original resolution, are ultimately aggregated or matched to this spatial framework.

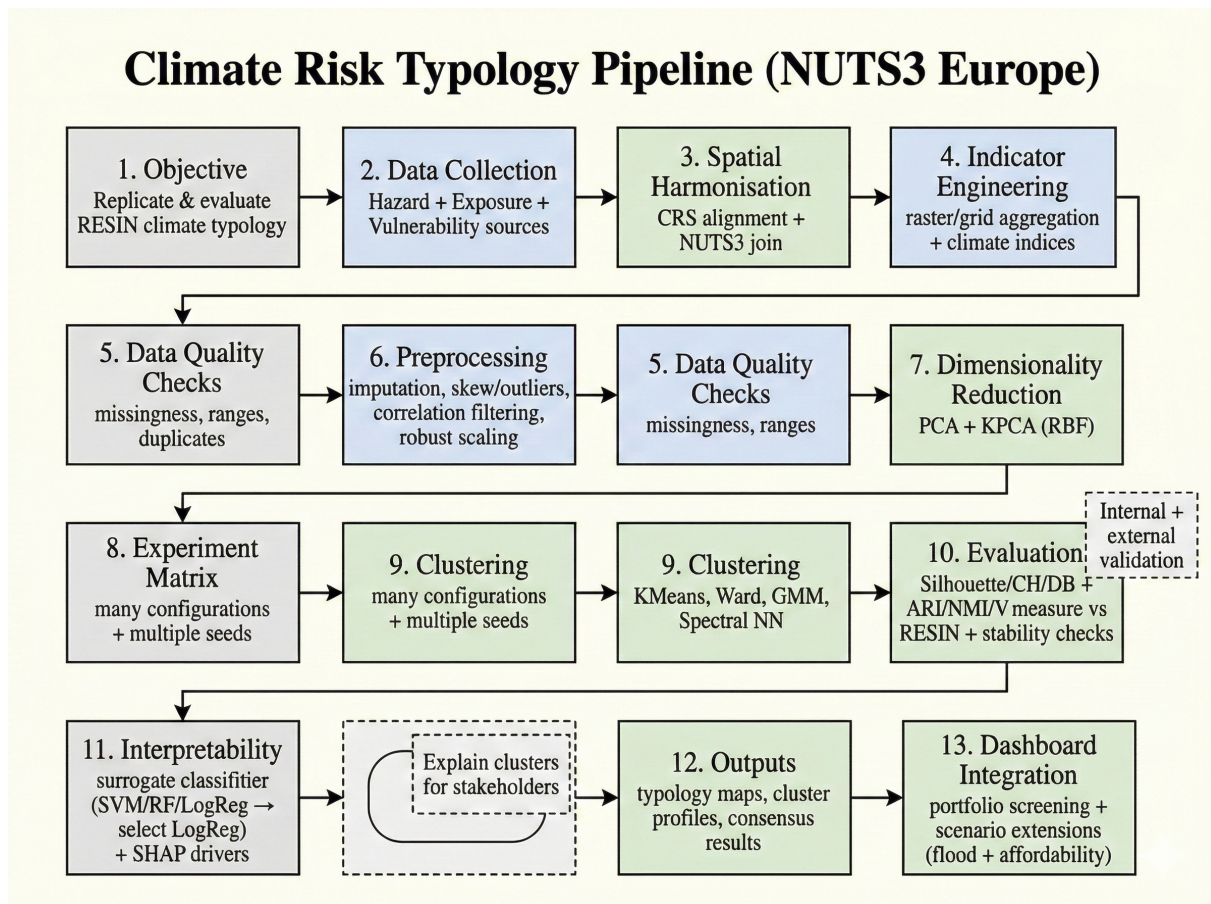
##### 5.1.1.1 Spatial Framework

The base geography is the official NUTS3 shapefile from Eurostat. Let  $\mathcal{R} = \{r_1, \dots, r_N\}$  denote the set of  $N$  regions with polygon geometries  $A(r)$  (area). All spatial datasets are reprojected to a common coordinate reference system (CRS) before overlay and aggregation. This allows consistent area-weighted statistics from rasters and reliable joins for tabular statistics.

##### 5.1.1.2 Climate Hazard Data

Physical climate hazards are represented using a combination of gridded climate projections and hazard-specific products. Main components include:

- **Temperature and precipitation indices:** daily temperature and precipitation projections from a EURO-CORDEX RCP 8.5 model at  $0.11^\circ$  resolution are used to construct standard indices such as mean/max temperature, summer days, heat waves, ice days, consecutive wet days and heavy precipitation days.
- **River flooding:** flood depth rasters for selected return periods are used to estimate potential fluvial flood hazard.



**Figure 5.1** – Overview of the climate risk typology pipeline at NUTS3 level. The workflow integrates heterogeneous hazard, exposure, and vulnerability data, applies robust preprocessing and dimensionality reduction, evaluates multiple clustering configurations, benchmarks results against expert typologies, and adds an interpretability layer using surrogate models and SHAP before delivery through an interactive dashboard.

- **Coastal flooding:** extreme water level information (tide and storm surge) is used to represent coastal inundation potential along the European coastline.
- **Drought:** a drought index (e.g. SPEI) is used to capture long-term moisture deficits.
- **Wildfire and landslide potential:** land-cover-based fire potential is derived from CORINE Land Cover and a landslide susceptibility product is used for mass-movement hazard in mountainous areas.

Each source is processed into one or more quantitative indicators per NUTS3 region and future time window.

### 5.1.1.3 Exposure Data

Exposure describes what is present in hazard-prone areas. We use:

- Land cover and built-up shares from CORINE Land Cover and related European datasets.
- Population counts and densities at NUTS3 level.
- Proxies for residential building stock / footprint where available.

### 5.1.1.4 Vulnerability Data

Vulnerability reflects how strongly people and systems are affected once a hazard occurs. The vulnerability block includes socio-economic and demographic indicators such as:

- Age structure (share of children and elderly).
- Income proxies / GDP per capita.
- Labour-market indicators (e.g. unemployment).
- Proxies for mobility and access to services (e.g. car ownership, urbanisation, access to care).

### 5.1.1.5 Reference Typology

For external benchmarking we use an expert-based European climate risk typology. These labels are *never* used during training; they serve only for ex-post comparison and validation of the data-driven partitions.

## 5.1.2 Indicator Construction

Raw datasets are transformed into a coherent indicator set grouped into three blocks: hazard ( $H$ ), exposure ( $E$ ), and vulnerability ( $V$ ). Let  $x_{r,j}$  denote indicator  $j$  for region  $r$ . The final feature matrix is  $X \in \mathbb{R}^{N \times p}$ .

### 5.1.2.1 Spatial Aggregation from Gridded Products

For a gridded variable  $z(s)$  defined over spatial locations  $s$ , we compute an area-weighted regional statistic:

$$\bar{z}_r = \frac{\int_{s \in r} z(s) ds}{\int_{s \in r} 1 ds} \approx \frac{\sum_{g \in \mathcal{G}(r)} z_g w_{rg}}{\sum_{g \in \mathcal{G}(r)} w_{rg}}, \quad (5.1)$$

where  $\mathcal{G}(r)$  is the set of grid cells intersecting region  $r$ ,  $z_g$  is the cell value, and  $w_{rg}$  is the area of intersection between region  $r$  and cell  $g$ . For additive quantities (e.g. exposed population in a mask), we use area-weighted sums instead of means.

For flood and coastal hazards, aggregation is restricted to land area where relevant (to avoid coastline artefacts).

### 5.1.2.2 Temporal Aggregation and Climate Indices

Daily climate variables are first transformed into annual indices (per grid cell) and then averaged over a target time slice. For example, summer days can be computed as:

$$SD_{g,y} = \sum_{d \in \mathcal{D}(y)} \mathbb{I}(T_{g,d}^{\max} > T_0), \quad (5.2)$$

where  $T_0$  is a threshold (e.g. 25°C),  $\mathcal{D}(y)$  is the set of days in year  $y$ , and  $\mathbb{I}(\cdot)$  is the indicator function. A time-slice mean is then:

$$SD_{g,\text{slice}} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} SD_{g,y}. \quad (5.3)$$

For change signals, we compute deltas relative to a baseline:

$$\Delta x_g = x_{g,\text{future}} - x_{g,\text{baseline}}. \quad (5.4)$$

All grid-level indices are then aggregated to NUTS3 using the spatial weighting above.

### 5.1.2.3 Block Structure and Directionality

Indicators are constructed to be monotone in risk interpretation: higher hazard values imply higher hazard, higher exposure values imply more assets/people at stake, and higher vulnerability values imply higher sensitivity or lower coping capacity. When a raw variable has the opposite meaning (e.g. income as adaptive capacity), we invert or re-code it so that the sign is consistent.

## 5.1.3 Preprocessing Pipeline

High-quality preprocessing is critical for meaningful clustering. The pipeline is implemented as a reproducible sequence in Dataiku and mirrored in Python notebooks.

### 5.1.3.1 Data Cleaning and Quality Checks

We verify:

- **Coverage:** region and country completeness.
- **Plausible ranges:** hard bounds for temperatures, proportions, non-negativity for counts.
- **Internal consistency:** shares within  $[0, 1]$ , totals within expected ranges.

Regions with systematic gaps in core hazard indicators are excluded from the clustering universe to avoid artefacts.

### 5.1.3.2 Missing Values

Missingness is handled conservatively. For low missing fractions, we impute using robust statistics, e.g. country-level medians:

$$x_{rj} \leftarrow \text{median}\{x_{r'j} : r' \in \text{country}(r), x_{r'j} \text{ observed}\}. \quad (5.5)$$

Indicators with poor coverage or unstable behaviour are dropped rather than forced into the feature space.

### 5.1.3.3 Outliers and Skewness

Many environmental and socio-economic indicators are heavy-tailed. We therefore apply either log transforms (where strictly positive) or winsorisation to reduce leverage of extremes. A typical winsorisation is:

$$x'_{rj} = \min(\max(x_{rj}, Q_j(\alpha)), Q_j(1 - \alpha)), \quad (5.6)$$

where  $Q_j(\cdot)$  is the empirical quantile function for feature  $j$  and  $\alpha$  is a small tail probability.

### 5.1.3.4 Correlation Filtering

To reduce redundancy and avoid overweighting a single process through near-duplicate variables, we remove one feature from pairs with high absolute correlation:

$$\text{if } |\rho(x_{\cdot j}, x_{\cdot k})| > \tau, \text{ drop either } j \text{ or } k, \quad (5.7)$$

with threshold  $\tau$  chosen based on exploratory analysis and interpretability considerations.

### 5.1.3.5 Robust Scaling

Clustering methods rely on distances, so features must be comparable in scale. We use robust scaling:

$$\tilde{x}_{rj} = \frac{x_{rj} - \text{median}(x_{\cdot j})}{\text{IQR}(x_{\cdot j})}, \quad (5.8)$$

where IQR is the interquartile range. This is less sensitive to outliers than z-score scaling.

Optionally, block-wise normalisation is tested so that hazard/exposure/vulnerability blocks have comparable influence.

## 5.1.4 Dimensionality Reduction

Dimensionality reduction is used to (i) reduce noise, (ii) mitigate multicollinearity, and (iii) obtain a stable representation for clustering.

### 5.1.4.1 Principal Component Analysis (PCA)

Given scaled data matrix  $\tilde{X}$ , PCA finds orthogonal directions that maximise variance. Formally, the first component loading  $w_1$  solves:

$$w_1 = \arg \max_{\|w\|=1} \text{Var}(\tilde{X}w), \quad (5.9)$$

with subsequent components orthogonal to previous ones. The PCA embedding is:

$$Z = \tilde{X}W, \quad (5.10)$$

where columns of  $W$  are the selected loadings. The number of components is chosen based on explained variance, stability of clustering, and interpretability.

### 5.1.4.2 Kernel PCA (KPCA) with RBF Kernel

Because the joint hazard–exposure–vulnerability space can contain non-linear structure, we also use kernel PCA. KPCA applies PCA in an implicit feature space via a kernel function. With an RBF kernel:

$$K_{ij} = \exp \left( -\gamma \|\tilde{x}_i - \tilde{x}_j\|^2 \right), \quad (5.11)$$

where  $\gamma$  controls the locality (smoothness) of the embedding. KPCA can capture curved manifolds and interactions between indicators that PCA cannot represent linearly. In our experiments, the selected reference solutions are based on a KPCA embedding with configuration KPCA\_RBF\_g0p001480.

### 5.1.5 Experiment Design and Model Selection

A key element of the study is systematic experimentation across the full modelling stack. Rather than selecting a single embedding or clustering method a priori, we evaluate a matrix of configurations and select reference solutions based on a combination of internal quality, external benchmarking, stability, and interpretability.

Concretely, we vary:

- **Representations:** PCA and KPCA (RBF) embeddings with multiple dimensionalities; where applicable we also test alternative non-linear embeddings (e.g., UMAP) for exploratory analysis.
- **Clustering algorithms:** k-means, Ward hierarchical clustering, GMM, and spectral clustering.
- **Hyperparameters:** number of components ( $d$ ), number of clusters ( $K$ ), and non-linear parameters such as the KPCA kernel scale  $\gamma$  (and neighbourhood choices for graph-based methods).
- **Stochasticity controls:** repeated runs with different random seeds (e.g., k-means initialisation) and, where relevant, multiple restarts to assess stability.

Each configuration is evaluated with the internal and external metrics defined in Section 5.1.7, together with practical degeneracy checks (cluster-size balance, avoidance of near-singleton clusters). Where multiple solutions are close in metric performance, preference is given to those with higher stability across reruns and clearer HEV-driven cluster narratives.

### 5.1.6 Clustering Techniques

Let  $z_r \in \mathbb{R}^d$  denote the reduced representation (PCA or KPCA) for region  $r$ .

#### 5.1.6.1 k-means

k-means partitions regions into  $K$  clusters by minimising within-cluster sum of squares:

$$\min_{\{C_k\}_{k=1}^K} \sum_{k=1}^K \sum_{r \in C_k} \|z_r - \mu_k\|^2, \quad (5.12)$$

where  $\mu_k$  is the centroid of cluster  $k$ . k-means is efficient and provides clear prototypes, but it assumes roughly convex clusters in the embedding space.

### 5.1.6.2 Agglomerative Hierarchical Clustering (Ward)

Ward's method merges clusters to minimise the increase in within-cluster variance. At each step it merges the pair  $(A, B)$  that minimises:

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2. \quad (5.13)$$

This produces a dendrogram that can be cut at different  $K$ .

### 5.1.6.3 Gaussian Mixture Models (GMM)

GMM assumes the data is generated by a mixture of Gaussians:

$$p(z) = \sum_{k=1}^K \pi_k \mathcal{N}(z \mid \mu_k, \Sigma_k), \quad (5.14)$$

with mixing weights  $\pi_k$ . Parameters are estimated by EM. GMM provides soft assignments and can model elliptical clusters but is sensitive to covariance assumptions and local optima.

### 5.1.6.4 Spectral Clustering

Spectral clustering uses a similarity graph. Given an affinity matrix  $A$  (e.g. RBF affinities), define the degree matrix  $D$  with  $D_{ii} = \sum_j A_{ij}$  and a graph Laplacian (one common form):

$$L_{\text{sym}} = I - D^{-1/2} A D^{-1/2}. \quad (5.15)$$

Clustering is then performed on the first  $K$  eigenvectors of  $L_{\text{sym}}$ , followed by k-means in that spectral embedding. Spectral methods can capture non-convex cluster geometry, but results depend on graph construction and neighbourhood parameters.

## 5.1.7 Evaluation Metrics

Cluster quality is assessed using internal metrics, external benchmarking against the expert typology, and practical interpretability checks.

### 5.1.7.1 Internal Validation

Let  $a(r)$  be the mean distance from  $r$  to other points in its cluster, and  $b(r)$  the minimum mean distance from  $r$  to points in any other cluster. The silhouette for region  $r$  is:

$$s(r) = \frac{b(r) - a(r)}{\max\{a(r), b(r)\}}. \quad (5.16)$$

We report the mean silhouette. We also compute Calinski–Harabasz and Davies–Bouldin indices to capture alternative compactness/separation trade-offs.

### 5.1.7.2 External Validation Against Expert Typology

Given algorithmic partition  $U$  and expert partition  $V$ , we compute:

- **ARI**: agreement corrected for chance.
- **NMI**: information overlap between partitions.

These metrics quantify overlap but do not imply one typology is “correct”; they indicate how much expert structure is recoverable from the indicator space alone.

### 5.1.7.3 Degeneracy and Business-Oriented Checks

Because large-scale geospatial datasets can produce misleadingly high internal scores (e.g. near-perfect silhouette from a dominant mega-cluster plus small outlier clusters), we apply additional checks:

- Cluster size balance (avoid solutions dominated by one/two clusters).
- Absence of singleton or near-singleton clusters unless clearly interpretable.
- Spatial coherence and stability under small perturbations.
- Interpretability of cluster profiles in hazard–exposure–vulnerability narratives.

### 5.1.8 Interpretability: Explaining Cluster Drivers with Surrogate Models and SHAP

Unsupervised clusters must be translated into defensible, human-readable narratives for investment and policy users. To explain *why* a region belongs to a given cluster, we adopt a two-step interpretability strategy:

1. Fit a supervised **surrogate classifier** that predicts the cluster label from the original scaled indicator space  $\tilde{X}$ . This converts the unsupervised output into a learnable mapping  $f : \tilde{X} \rightarrow \hat{c}$ .
2. Apply **SHAP** to the surrogate model to quantify feature contributions to each cluster prediction and to extract stable global drivers for cluster membership.

#### 5.1.8.1 Surrogate Model Selection

We evaluate multiple candidate surrogates—linear SVM, logistic regression, and random forest—using repeated cross-validation on the same feature set used for clustering. While random forests can capture non-linear interactions, they can also overfit and produce less stable global narratives across reruns. In our experiments, **logistic regression** provided the best balance of (i) predictive performance on held-out folds and (ii) interpretability through linear coefficients and stable SHAP attributions. Therefore, logistic regression is selected as the reference surrogate.

#### 5.1.8.2 SHAP Attribution for Cluster Narratives

Given the trained surrogate model, SHAP values are computed for each feature and each region. For a given class/cluster, we summarise:

- **Global drivers:** features with the largest mean absolute SHAP contribution for that cluster.
- **Directional interpretation:** whether high feature values increase or decrease membership likelihood.
- **Cluster profiles:** combining SHAP summaries with raw/standardised feature contrasts to create an HEV narrative (hazard-dominant vs. exposure-dominant vs. vulnerability-dominant signatures).

This interpretability layer is used to (i) label clusters in an evidence-based way, (ii) verify that discovered types are not artefacts of scaling or outliers, and (iii) communicate results to non-technical stakeholders.

### 5.1.9 Sensitivity and Scenario Testing

We test robustness to key modelling decisions.

### 5.1.9.1 Indicator and Block Sensitivity

We compare hazard-only vs. full H-E-V feature sets, and test alternative feature subsets and block scaling choices. Stability is quantified via agreement between clusterings (e.g. ARI between solutions) and by map-based inspection of stable vs. unstable regions.

### 5.1.9.2 Methodological Sensitivity

We compare:

- PCA vs. KPCA embeddings,
- k-means vs. hierarchical vs. GMM vs. spectral clustering for comparable  $K$ ,
- multiple random initialisations where applicable.

### 5.1.9.3 Scenario Interpretation

Although the analysis focuses on a high-emission climate scenario, the pipeline is designed to be re-runnable for alternative scenarios, time windows and updated hazard products. Cluster profiles are interpreted in a forward-looking way, highlighting how regions may transition between risk types under warming.

In summary, the methodology combines careful indicator engineering with a transparent unsupervised learning framework. Robust preprocessing and defensible model selection criteria ensure that the resulting typology is driven by meaningful climate-risk structure rather than artefacts of scale, outliers or imbalanced partitions. The added interpretability layer (surrogate model + SHAP) links clusters back to the original HEV indicators, enabling clear narratives and business adoption.

# Chapter 6

## Results

### 6.1 Results

This chapter presents the main empirical results of the study. We first discuss the coverage and behaviour of the constructed indicators, then analyse the structure of the climate feature space, followed by the outcomes of the clustering experiments and their comparison with the expert typology. Finally, we show how these results translate into composite climate risk scores and summarise robustness checks and model selection.

#### 6.1.1 Data Coverage and Indicator Behaviour

After data cleaning and harmonisation (see Section 5.1.3), we obtain a consolidated dataset covering the large majority of European NUTS3 regions with complete or near-complete information on the selected hazard, exposure and vulnerability indicators.

Only a small subset of regions had to be removed from the clustering universe because of systematic gaps in key indicators (for example, missing flood depth data or incomplete socio-economic statistics). For the remaining regions, missing values are limited and handled through controlled imputation. This means that the final feature matrix is both dense and geographically well-distributed, without large clusters of missing data in specific countries.

Initial descriptive analysis shows that:

- Hazard indicators such as mean temperature change, number of summer days and heavy precipitation days exhibit clear large-scale gradients, with warmer and drier conditions in the southern Mediterranean and cooler, wetter conditions in the north-western Atlantic regions.
- Flood and coastal indicators are strongly concentrated along major river basins and low-lying coastlines, as expected.
- Exposure indicators (population density, built-up shares) have a highly skewed distribution: a small number of metropolitan regions account for a very large share of total exposed population.
- Vulnerability indicators (age structure, income, labour market) vary more smoothly across space, with clusters of high vulnerability often appearing in economically weaker or more peripheral regions.

From a purely descriptive point of view, the constructed indicators behave in a way that is consistent with domain knowledge and with the underlying data sources, which gives confidence that the preprocessing pipeline did not introduce obvious artefacts.

### 6.1.2 Structure of the Climate Feature Space

To understand the dominant patterns in the combined hazard–exposure–vulnerability space, we first apply Principal Component Analysis (PCA) to the robustly scaled indicators as a diagnostic tool.

Across experiments, the first few components capture the main continental-scale gradients:

- **PC1** is largely dominated by thermal and drought-related indicators. High PC1 scores typically correspond to warmer, drier regions and more frequent heat extremes; low PC1 scores correspond to cooler, wetter climates.
- **PC2** is driven mainly by precipitation extremes and fluvial/coastal flood indicators, separating flood-prone river basins and coastal plains from more inland or elevated regions.
- **PC3** combines socio-economic vulnerability with exposure. High scores are associated with higher shares of elderly population, weaker labour-market outcomes and lower income proxies, often combined with non-trivial exposed population.

While PCA is useful for interpretability, it is strictly linear. We therefore also explore non-linear representations using **kernel PCA (KPCA) with an RBF kernel**. The KPCA embedding is better suited for capturing curved structure and interactions between indicators (for example, non-linear combinations of hazard intensity and socio-economic coping capacity). The selected reference solutions in this thesis are built on a KPCA embedding with parameter `KPCA_RBF_g0p001480`.

### 6.1.3 Clustering Results

We experiment with several clustering algorithms and a range of cluster numbers. Across methods, a medium-granularity solution (single-digit number of clusters) offers the best compromise between interpretability, spatial coherence, and robustness.

#### 6.1.3.1 Internal Cluster Quality and the “Too-Good-to-Be-True” Problem

Internal validation metrics (silhouette, Calinski–Harabasz, Davies–Bouldin) are informative, but in this setting they must be interpreted carefully.

In large, high-dimensional regional datasets, it is possible to obtain *extremely high* silhouette values that are *not useful* for a typology. In our grid search, several PCA-based solutions produced silhouette scores close to 1.0. Manual inspection revealed that these runs were often **degenerate partitions** dominated by one very large cluster (absorbing most regions) and several tiny clusters or singletons. Such solutions inflate separation metrics because outliers become isolated “clusters” and the majority becomes a single compact group, but this does not yield a meaningful multi-type European typology.

Because the goal is to **challenge and replicate** the RESIN typology (which uses eight classes), we introduced an additional technical screening step beyond internal indices:

- Exclude solutions with **highly imbalanced cluster-size distributions** (one or two dominant clusters).
- Exclude solutions producing **singleton / near-singleton clusters** that are not stable or interpretable.
- Prefer solutions that are **spatially coherent** (neighbouring regions tend to share types) and interpretable in terms of hazard–exposure–vulnerability narratives.

This screening is essential in practice: it prevents selecting models that look excellent numerically but fail as a portfolio-relevant typology.

### 6.1.3.2 Comparison with Expert Typology

We compare the data-driven clusters with the expert-based regional climate typology using external validation metrics such as the Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI).

Absolute values are expected to be modest because the expert typology and our indicator set are not identical. However, external metrics are clearly above chance for the short-listed configurations, and (more importantly) the spatial patterns align well: Mediterranean heat/drought regions, flood belts along major basins, and northern cool/wet regions emerge consistently.

Most disagreements occur in transition zones (between climate regimes) and in regions where socio-economic vulnerability plays a stronger role in our multi-block feature space than in the expert typology.

### 6.1.3.3 Spatial Coherence and Map-Level Agreement

Quantitative external metrics (ARI/NMI) provide one lens on agreement with the expert typology, but for a European regional typology the **map-level structure** is equally important: a useful typology should yield spatially coherent, interpretable regions rather than fragmented “salt-and-pepper” patterns.

Figure 6.1 compares the expert typology map to the two selected data-driven reference solutions. At the macro scale, both algorithmic solutions reproduce the main continental patterns that are expected from climate processes and geography: a Mediterranean heat/drought regime, northern cool/wet regions, and flood-related belts associated with major basins and low-lying/coastal areas. This provides strong evidence that the clustering is capturing real, large-scale structure in the hazard–exposure–vulnerability feature space rather than artefacts.

The differences between the two models are also informative. The **KPCA + KMeans** ( $K = 8$ ) map is deliberately aligned to the RESIN class count and therefore acts as a like-for-like typology benchmark. The **KPCA + Spectral** ( $K = 10$ ) solution preserves the same broad structure but introduces a finer segmentation in several transition zones, effectively splitting some broader types into sub-types with more homogeneous combinations of hazards and/or vulnerability. In practice, these extra splits are useful as a robustness lens: they indicate where the  $K = 8$  types contain meaningful internal heterogeneity.

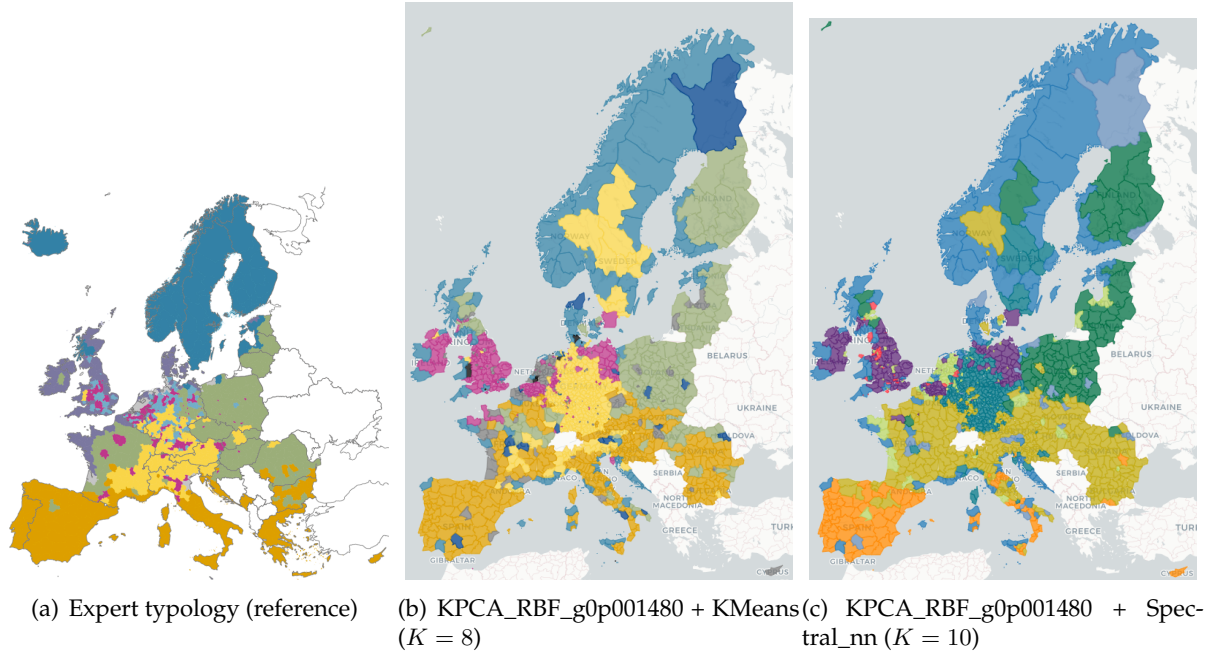
## 6.1.4 Interpretability and Driver Analysis with SHAP

A clustering typology is only useful for decision support if we can explain *why* regions end up in a given type. Purely unsupervised labels do not provide this by default: the algorithm returns assignments, but not drivers. To address this, we added an interpretability layer based on **SHAP (SHapley Additive exPlanations)**, which is a modern, model-agnostic framework for explaining predictions using concepts from cooperative game theory.

### 6.1.4.1 Why SHAP

Traditional interpretability approaches each have clear limitations:

- **Cluster centroids / means** describe what a cluster looks like on average, but they do not explain which features *caused* an individual region to be assigned to that cluster.
- **Correlation-based explanations** can be misleading in multi-collinear indicator spaces and do not capture non-linear effects or interactions.
- **Permutation importance** is global and unstable under correlated features; it also does not provide a consistent local explanation for specific regions.



**Figure 6.1** – Map comparison between the expert typology and the two selected data-driven reference solutions on the NUTS3 grid. Both algorithmic maps recover the same macro-regimes as the expert typology, while spectral clustering provides additional sub-structure in transition zones.

SHAP addresses these issues by producing **local explanations** (per region) that add up consistently to a model output, and by aggregating them into **global importance** patterns. In practice, SHAP provides both: (i) a ranked list of the most influential indicators overall, and (ii) cluster-specific driver signatures that can be translated into climate-risk narratives.

#### 6.1.4.2 Method: Surrogate Model + SHAP

Because clustering itself does not have a natural “prediction function” like a supervised model, we follow a common interpretability strategy: we train a **supervised surrogate classifier** to predict the final cluster labels from the original hazard–exposure–vulnerability indicators (after the same preprocessing). The surrogate is used only for interpretation; it does not change the cluster assignments.

Let  $y_i \in \{1, \dots, K\}$  be the cluster label for region  $i$ , and  $\mathbf{x}_i \in \mathbb{R}^p$  the vector of preprocessed indicators. We train a classifier  $f(\mathbf{x})$  such that:

$$\hat{y}_i = f(\mathbf{x}_i).$$

SHAP then decomposes the model output into feature contributions. For a given class  $c$ , the SHAP explanation for an instance  $i$  can be written as:

$$f_c(\mathbf{x}_i) = \phi_{0,c} + \sum_{j=1}^p \phi_{j,c}(\mathbf{x}_i),$$

where  $\phi_{j,c}(\mathbf{x}_i)$  is the contribution of feature  $j$  to the model score for class  $c$ , and  $\phi_{0,c}$  is the baseline (expected) score. Positive SHAP values push a region *towards* class  $c$ , negative values push it *away*.

### 6.1.4.3 Global and Cluster-Specific Drivers

Figure 6.2 shows SHAP summaries for the spectral typology, highlighting the strongest drivers for cluster membership. The plots show that cluster assignment is not determined by a single variable, but by consistent **multi-indicator signatures** that align with the hazard–exposure–vulnerability framework.

In particular, the SHAP patterns confirm that:

- Heat and drought-related indicators dominate the separation of southern and Mediterranean types.
- Flood/coastal hazard indicators drive the identification of river-basin and coastal low-land types.
- Exposure variables (population, built-up proxies) differentiate metropolitan high-consequence regions even when physical hazards are moderate.
- Vulnerability indicators (age structure, labour market and income proxies) are key for identifying regions where coping capacity is structurally lower, shifting them into higher-risk narratives under the same hazard level.

### 6.1.4.4 Why this matters for the thesis

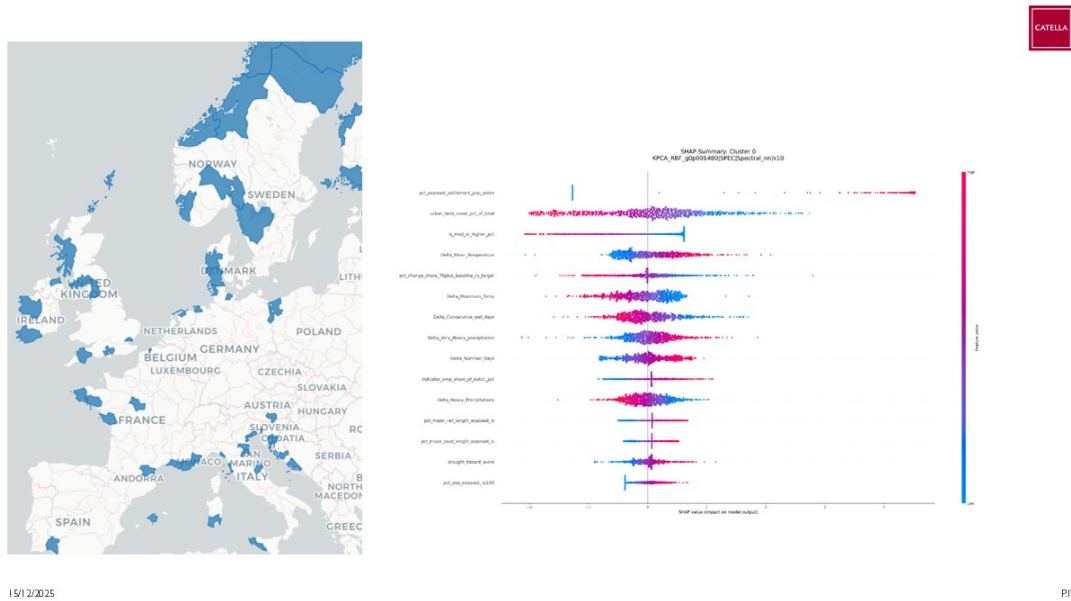
This interpretability layer is more than a visual add-on: it is the mechanism that connects the typology to portfolio-relevant reasoning. Instead of saying “the algorithm grouped these regions together”, we can say *which hazard, exposure, and vulnerability signals* made that grouping happen, and whether the drivers are consistent with climate-risk understanding. This is essential for transparency, for stakeholder trust, and for using the typology as a decision-support tool rather than a black-box clustering output.

### 6.1.4.5 Regional Cluster Profiles

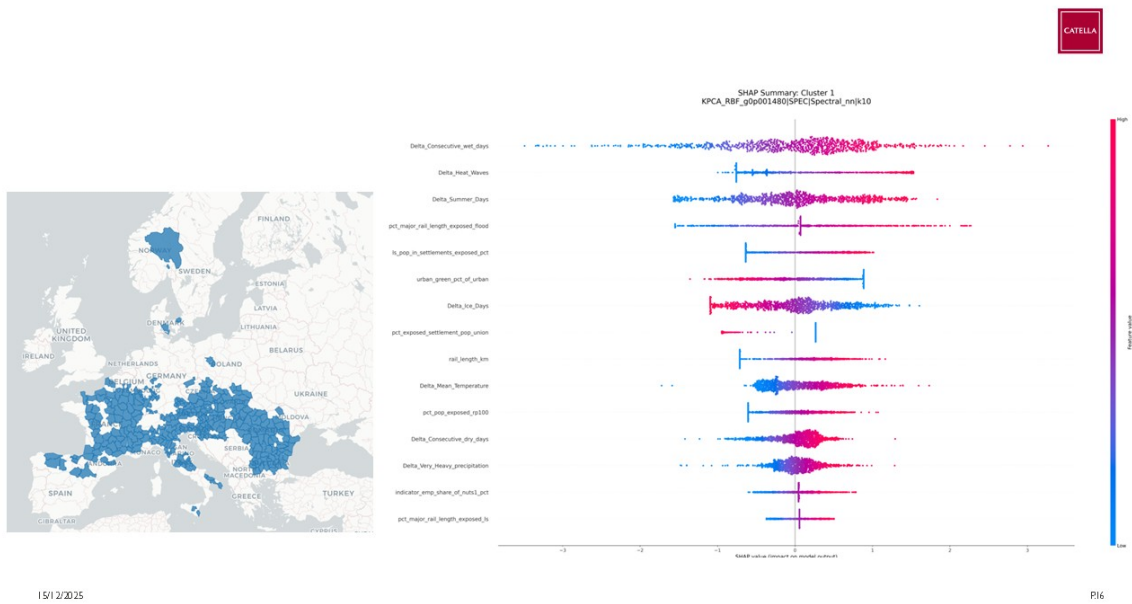
For interpretation, we compute average indicator values per cluster and assign simple descriptive names. A typical structure looks as follows:

- **Hot and Dry Southern Regions:** High heat and drought indicators, relatively lower river flood hazard, mixed exposure, and moderate to high vulnerability. Many Mediterranean inland regions fall into this type.
- **Coastal and River Flood Belts:** Elevated fluvial and coastal indicators, moderate temperatures, and significant exposed population and built-up area in hazard-prone zones.
- **Highly Exposed Metros:** Moderate physical hazard but very high exposure due to dense urban population and large residential building stock; vulnerability is typically lower, partially compensating exposure.
- **Peripheral, High-Vulnerability Regions:** Moderate hazards, relatively low exposure, but higher vulnerability (ageing, weaker labour markets, lower incomes), implying lower coping capacity.
- **Mountain / Upland Regions:** Lower exposure, mixed socio-economic profiles, and more localised hazard combinations (e.g. wildfire/landslide proxies), often forming distinct types.

Mapping these clusters back to Europe produces coherent geographic patterns. From an investment perspective, the typology compresses a high-dimensional space into a small set of portfolio-relevant climate risk “stories”.



(a) SHAP summary for spectral typology (cluster view 0)



(b) SHAP summary for spectral typology (cluster view 1)

**Figure 6.2** – SHAP-based interpretability of the surrogate model trained to reproduce the spectral clustering labels. The plots highlight both global and cluster-specific driver patterns, enabling translation from algorithmic labels to hazard–exposure–vulnerability narratives.

#### 6.1.4.6 Clustering Performance and Model Selection

To construct a robust climate risk typology, we evaluated a grid of clustering algorithms and cluster numbers  $K$  on the robustly scaled hazard–exposure–vulnerability feature space, using both PCA and KPCA embeddings. The goal was to identify solutions that: (i) have acceptable internal structure, (ii) show non-trivial agreement with the expert-based RESIN typology, and (iii) produce spatially coherent, interpretable regional types.

The model grid included k-means, agglomerative clustering (Ward and average linkage), Gaussian mixture models, and spectral clustering. For partitioning methods, we explored  $K$  between 4 and 12 and ran multiple random initialisations where applicable.

Critically, we did *not* select models purely by maximising silhouette. Instead, we used a combined selection logic:

- remove degenerate solutions (dominant clusters / singletons),
- retain solutions that give meaningful partitions and stable spatial patterns,
- then compare internal metrics and (where available) external agreement with RESIN.

Table 6.1 summarises the internal metrics for the two solutions selected as reference models: **KPCA\_RBF\_g0p001480 + KMeans** ( $K = 8$ ) as the main typology (chosen for comparability with RESIN and strong practical separation with balanced cluster sizes), and **KPCA\_RBF\_g0p001480 + Spectral\_nn** ( $K = 10$ ) as a more fine-grained alternative with stronger external agreement.

**Table 6.1** – Internal validation for the two reference clustering solutions.

| Algorithm                        | $K$ | Silhouette | Calinski–Harabasz | Davies–Bouldin |
|----------------------------------|-----|------------|-------------------|----------------|
| KPCA_RBF_g0p001480 + KMeans      | 8   | 0.2745     | 1 100.62          | 1.0474         |
| KPCA_RBF_g0p001480 + Spectral_nn | 10  | 0.1934     | 643.55            | 1.3912         |

**6.1.4.6.1 Why the KPCA + KMeans ( $K = 8$ ) solution is selected despite lower silhouette than “perfect” PCA runs.** The key technical reason is that the PCA runs with silhouette close to 1 were typically **not true eight-type segmentations**. They were often dominated by a single macro-cluster plus tiny outlier clusters, which is not suitable for a typology intended for interpretation and portfolio screening. In contrast, the selected KPCA + KMeans ( $K = 8$ ) solution produces a **balanced partition** (largest cluster  $\approx 27\%$  of regions; smallest cluster is small but not a singleton), and yields stable, interpretable geographic patterns consistent with the hazard–exposure–vulnerability logic.

**6.1.4.6.2 High- $k$  validation.** To study the high- $k$  region of the grid in more detail, we performed a dedicated validation experiment for configurations with  $k = 8$ –10. The setup is summarised in Table 6.2.

**Table 6.2** – Overview of the validation setup for high- $k$  clustering runs ( $k = 8$ –10).

| Category               | Outcome                                    |
|------------------------|--------------------------------------------|
| Evaluated runs         | 10 (all valid)                             |
| IDs matched with RESIN | 1 386 total; 1 304 used per run            |
| Consensus $K^*$        | 19 clusters (from co-association)          |
| Winner configuration   | KPCA_RBF_g0p001480 + Spectral_nn, $k = 10$ |
| Stability matrix       | $10 \times 10$ (ARI between runs)          |

Table 6.3 reports internal and external metrics for the shortlisted high- $k$  runs used. ARI and NMI describe agreement with the RESIN typology, while Silh., DB and CH report internal compactness and separation.

**Table 6.3** – Internal and external validation metrics for shortlisted high- $k$  clustering runs.

| Embedding          | Stage | Algorithm             | $k$ | ARI    | NMI    | Silh.  | DB     | CH         | $n_{\text{match}}$ | Rank    |
|--------------------|-------|-----------------------|-----|--------|--------|--------|--------|------------|--------------------|---------|
| PCA_k14            | KM    | KMeans                | 8   | 0.0202 | 0.0794 | 0.9796 | 0.3892 | 179 538.18 | 1 304              | 1.0395  |
| PCA_k14            | AGG   | Agglomerative_average | 10  | 0.0202 | 0.0804 | 0.9794 | 0.3972 | 227 921.53 | 1 304              | 1.0359  |
| PCA_k14            | AGG   | Agglomerative_ward    | 8   | 0.0212 | 0.0832 | 0.9786 | 0.4054 | 153 472.92 | 1 304              | 1.0333  |
| PCA_k14            | AGG   | Agglomerative_ward    | 10  | 0.0212 | 0.0840 | 0.9788 | 0.4245 | 216 785.57 | 1 304              | 1.0244  |
| KPCA_RBF_g0p001480 | SPEC  | Spectral_nn           | 10  | 0.1881 | 0.3187 | 0.1934 | 1.3912 | 643.55     | 1 304              | 0.0399  |
| KPCA_RBF_g0p001480 | GMM   | GMM_full              | 8   | 0.1492 | 0.2346 | 0.2262 | 1.3520 | 821.33     | 1 304              | 0.0113  |
| KPCA_RBF_g0p044398 | SPEC  | Spectral_nn           | 8   | 0.1769 | 0.3147 | 0.2089 | 1.4860 | 204.79     | 1 304              | -0.0052 |
| KPCA_RBF_g0p044398 | SPEC  | Spectral_nn           | 10  | 0.1610 | 0.3094 | 0.2004 | 1.4465 | 202.33     | 1 304              | -0.0125 |
| KPCA_RBF_g0p001480 | GMM   | GMM_full              | 10  | 0.1547 | 0.2594 | 0.1880 | 1.3865 | 743.06     | 1 304              | -0.0263 |
| KPCA_RBF_g0p001480 | GMM   | GMM_diag              | 10  | 0.1547 | 0.2594 | 0.1429 | 1.5148 | 570.93     | 1 304              | -0.1355 |

Table 6.4 highlights the two configurations that we carry forward as reference models in the remainder of the thesis: **KPCA + KMeans** ( $K = 8$ ) as the main RESIN-comparable typology, and **KPCA + Spectral\_nn** ( $K = 10$ ) as a non-linear, more fine-grained alternative with stronger external agreement.

**Table 6.4** – Comparison of the two final reference configurations used in the thesis.

| Embedding          | Algorithm   | Type                    | $k$ | ARI    | NMI    | Silh.  | DB     | CH       |
|--------------------|-------------|-------------------------|-----|--------|--------|--------|--------|----------|
| KPCA_RBF_g0p001480 | KMeans      | RESIN-comparable (main) | 8   | 0.2112 | 0.3022 | 0.2745 | 1.0474 | 1 100.62 |
| KPCA_RBF_g0p001480 | Spectral_nn | Non-linear (robustness) | 10  | 0.1881 | 0.3187 | 0.1934 | 1.3912 | 643.55   |

**6.1.4.6.3 External validation against RESIN.** To quantify overlap with the expert-based European Climate Risk Typology, we treat the RESIN classes as an external reference and compute ARI and NMI for shortlisted candidate solutions (where matching IDs are available). The strongest external agreement in the high- $k$  shortlist is achieved by the KPCA\_RBF\_g0p001480 + Spectral\_nn configuration with  $k = 10$  (Table 6.3), indicating that the non-linear embedding combined with a graph-based clustering method recovers a larger share of expert structure.

At the same time, we retain an explicit  $k = 8$  solution for direct comparability with RESIN. The selected KPCA + KMeans ( $k = 8$ ) typology is therefore used as the main “like-for-like” reference throughout the thesis, while the spectral solution provides a complementary, higher-resolution view.

**6.1.4.6.4 Selected Solutions.** Based on the combination of internal evidence, external agreement (where available), and visual inspection of maps and indicator profiles, we select two reference solutions:

1. **KPCA\_RBF\_g0p001480 + KMeans with  $K = 8$**  as the main typology used throughout the thesis. This has three advantages: (i) it offers a direct, fair comparison to RESIN (which also uses eight classes); (ii) it avoids degenerate “too-good-to-be-true” solutions by producing a balanced partition; and (iii) the eight clusters can be translated into clear climate risk narratives aligned with the hazard–exposure–vulnerability framework.
2. **KPCA\_RBF\_g0p001480 + Spectral\_nn with  $K = 10$**  as an alternative, more fine-grained view. This solution preserves the same macro-structure but splits several types into more homogeneous subtypes and shows stronger quantitative overlap with the RESIN typology in external validation.

The two typologies are not contradictory; rather, they offer two levels of granularity built on the same underlying data and preprocessing pipeline. In the remainder of the thesis, KPCA

+ KMeans ( $K = 8$ ) is used as the default for interpretation and mapping, while KPCA + Spectral\_nn ( $K = 10$ ) is used as a robustness check and for highlighting sub-patterns within broader regional types.

### 6.1.5 Composite Climate Risk Scores

Using the hazard, exposure and vulnerability indicators, we compute a composite climate risk score for each region, following the weighting strategy described in Section 5.1.2 and Section 5.1.3.

The distribution of scores is continuous, with a long upper tail representing regions that combine high hazard with either high exposure, high vulnerability, or both. When we overlay the composite score with the cluster labels, we find that:

- Clusters dominated by extreme hazards and high exposure (e.g. coastal flood belts, dense urban heat centres) systematically occupy the upper part of the score distribution.
- Clusters with moderate hazards but high vulnerability are placed in the middle to upper-middle part of the distribution, reflecting that socio-economic fragility matters even when hazards are not extreme.
- Low-hazard, low-exposure regions form the lower end of the distribution. These regions may still be relevant for local risk management, but from a pan-European portfolio perspective they represent comparatively lower climate-related pressure.

The composite score and the clusters are strongly but not perfectly correlated. This is useful: the score provides a simple ranking of regions, while the clusters provide a richer typology that separates regions with similar overall risk level but different underlying drivers.

### 6.1.6 Sensitivity and Robustness of the Results

We run a set of sensitivity experiments to test how robust the results are to key modelling choices:

- **Hazard-only vs. full feature set:** When clustering is performed on hazard indicators alone, the solution is dominated by broad north–south and coastal–inland gradients. Adding exposure and vulnerability splits several of these groups into more meaningful subtypes (for example, separating wealthy but exposed coastal regions from poorer, less-prepared ones).
- **Alternative scaling and outlier treatment:** Using standard z-score scaling instead of robust scaling makes the clustering more sensitive to extreme regions and reduces stability. The robust pipeline yields more coherent and interpretable types.
- **Different algorithms and  $K$ :** Across k-means, hierarchical clustering, GMMs and spectral clustering, the same macro-patterns remain visible (Mediterranean heat/drought, flood belts, high-vulnerability peripheries). Changing  $K$  mainly affects how finely these macro-patterns are split.

Overall, the macro structure of the typology is stable across reasonable choices of method and hyperparameters. This gives confidence that the main findings are not artefacts of a single algorithmic choice, but reflect real, large-scale patterns in the underlying data.

In summary, the results show that it is possible to construct a coherent, data-driven climate risk typology for European NUTS3 regions using openly available indicators on hazards, exposure and vulnerability. The unsupervised clusters reproduce a meaningful part of an existing expert typology while also revealing new distinctions that are important from an investment and resilience point of view.

### 6.1.6.1 Surrogate Classifier Comparison (SVM vs. Random Forest vs. Logistic Regression)

Because SHAP explanations depend on the underlying predictive model, we tested multiple surrogate classifiers for reconstructing the cluster labels from the preprocessed hazard–exposure–vulnerability feature space. Concretely, we trained and evaluated: (i) a linear SVM, (ii) a Random Forest classifier, and (iii) multinomial Logistic Regression. All models were trained on the same input features used for clustering (after identical preprocessing), and evaluation was performed using repeated stratified cross-validation to ensure that results were not driven by a single split.

Across these experiments, **Logistic Regression** provided the best overall balance between (a) predictive fidelity to the cluster labels and (b) interpretability stability. While Random Forest and SVM models can achieve competitive predictive performance, we observed two practical limitations for explanation quality in this setting:

- **Random Forest:** SHAP explanations tend to be more fragmented across correlated indicators and can vary more across reruns because tree-based models distribute importance across interacting splits. In a high-dimensional HEV feature space with substantial multicollinearity, this can produce noisier global narratives (many features with small, unstable contributions) even when classification accuracy is high.
- **SVM:** although linear SVMs can fit the labels reasonably well, explanation pipelines are less direct (especially when probability calibration is required), and SHAP-style attributions can become more sensitive to scaling and class imbalance. This reduced the clarity of cluster-level driver summaries compared to the logistic regression surrogate.

In contrast, the Logistic Regression surrogate yields **stable, sparse, and directionally consistent** driver signatures. Practically, this makes the SHAP summaries easier to communicate to non-technical stakeholders: features that increase membership in a cluster do so in a consistent direction, and the global ranking of drivers is less sensitive to reruns. For this reason, all SHAP plots reported in this thesis are based on the Logistic Regression surrogate unless stated otherwise.

# Chapter 7

## Discussion

### 7.1 Discussion

This chapter interprets the empirical results from Section 6.1 and links them back to the goal of the thesis: building a scalable, data-driven climate risk typology for European NUTS3 regions that is useful for investment screening and portfolio risk communication.

The discussion focuses on (i) what the clusters represent in climate-risk terms, (ii) why our data-driven typologies look similar to the expert-based RESIN typology at large scale, (iii) where and why they differ, and (iv) what the comparison between **KPCA + k-means** ( $K = 8$ ) and **KPCA + spectral clustering** ( $K = 10$ ) reveals about the structure of the data.

#### 7.1.1 What the Clusters Represent in Climate-Risk Terms

Across all experiments, the European NUTS3 landscape naturally breaks into a small number of recurring climate-risk *stories*. These are not abstract mathematical groupings; they reflect recognisable combinations of **hazard**, **exposure**, and **vulnerability**:

- **Heat–drought dominated regions:** mainly southern and Mediterranean areas with stronger warming, more frequent heat extremes, and elevated drought pressure.
- **Flood-driven lowlands and corridors:** river basins and coastal lowlands where fluvial/coastal signals dominate and risk is amplified by exposed settlements and infrastructure.
- **High-exposure metropolitan regions:** moderate hazards in absolute terms, but very high consequences because dense population, transport, and built-up assets concentrate the exposure.
- **High-vulnerability regions:** areas with moderate hazards but limited coping capacity (ageing, labour market weakness, lower income proxies), which pushes risk upwards even without extreme hazards.
- **Topography- and land-cover-driven regions:** upland and mixed terrain areas where landslide/water-stress proxies, land cover and local exposure patterns create distinct profiles.

From a real-estate perspective, this is exactly the compression that is needed: a high-dimensional indicator space is translated into a set of interpretable categories that can be discussed with non-technical stakeholders.

#### 7.1.2 Why Our Typologies Look Similar to RESIN at Macro Scale

A key result is that our typologies reproduce the same broad European structure that is visible in RESIN maps: southern heat/drought zones, lowlands and flood corridors, north-western urban belts, and northern/green transition regions.

This convergence is expected for three reasons:

- **Shared physical drivers:** temperature, precipitation, coastal exposure, and flood probability follow large-scale climate and geographic gradients. Any indicator set that measures these processes will recover similar spatial patterns.
- **Comparable spatial unit:** RESIN and this thesis both operate at NUTS3 level. This matters because aggregation tends to highlight continental-scale structure and suppress some local noise.
- **A physically constrained feature design:** our pipeline uses interpretable hazard indices and exposure proxies rather than arbitrary transformations. This pushes clustering toward meaningful climate regimes rather than artefacts.

In short: the algorithms are not discovering a different climate system. They are re-assembling the same underlying signals, but in a transparent and reproducible way.

### 7.1.3 Where and Why We Differ from RESIN (and Why That is Useful)

Differences between our clusters and RESIN are concentrated in three settings:

- **Transition zones:** boundary regions between Atlantic–continental or Mediterranean–temperate regimes. Here small shifts in indicator balance move regions across cluster boundaries.
- **Socio-economic re-weighting:** RESIN’s narrative is often more hazard-forward, whereas our feature set gives vulnerability and exposure more space. This pulls some moderate-hazard regions into higher-risk stories because coping capacity is low or exposure is high.
- **Data coverage and measurement:** where some indicators are uncertain or inconsistent across countries, a rule-based expert label may differ from a purely data-driven label that follows the available inputs strictly.

These disagreements are not “errors”; they are the most informative regions for investors. They flag where (i) exposure dominates hazard, (ii) vulnerability amplifies otherwise moderate hazards, or (iii) more due diligence is needed due to data limitations.

### 7.1.4 Why k-means and Spectral Work (and Why DBSCAN Fails Here)

#### 7.1.4.1 Why DBSCAN (and similar density methods) fail in this setting

DBSCAN is attractive because it can find irregular cluster shapes and detect noise. However, in our problem it performed poorly and often produced degenerate outputs (one large cluster + noise, or almost everything labelled as noise). The reasons are structural:

- **Distance concentration in high dimensions:** even after scaling, nearest-neighbour distances tend to become less informative as dimensionality increases. Density thresholds become unstable.
- **Non-uniform sampling density across Europe:** NUTS3 regions are not distributed as a uniform “cloud” in feature space. Metropolitan outliers and rare hazard combinations break the assumption of consistent density.
- **Parameter sensitivity:** small changes in  $\epsilon$  or `min_samples` flip results from “all noise” to “one cluster”, which is not usable for a stable typology.

So DBSCAN fails not because it is a bad method, but because density assumptions do not match the geometry of this hazard–exposure–vulnerability dataset.

#### 7.1.4.2 Why k-means and Spectral\_nn succeed

Our best-performing solutions use **kernel PCA** first and then apply either a partitioning method (k-means) or a graph partition method (spectral). This is a strong combination for two reasons:

- **Kernel PCA reshapes the geometry:** it separates non-linear interactions between indicators (e.g., hazard intensity interacting with exposure and socio-economic capacity).
- **The clustering objective matches the embedding:** k-means performs well when clusters are roughly compact in the embedding, while spectral clustering performs well when clusters are best separated by connectivity in a similarity graph rather than by straight-line boundaries.

#### 7.1.5 Why the RBF KPCA Setting (g0p001480) Performs Well

Kernel PCA with an RBF kernel uses:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2).$$

The parameter  $\gamma$  controls the length scale of similarity:

- If  $\gamma$  is **too small**, the kernel becomes almost constant and KPCA behaves close to linear PCA (little benefit).
- If  $\gamma$  is **too large**, only extremely close neighbours remain similar; the embedding becomes overly fragmented and clustering becomes unstable (overfitting).

The selected value `KPCA_RBF_g0p001480` sits in the useful middle regime: it is strong enough to separate non-linear regimes (compound flood/heat, transport-exposure corridors, urban land-cover transitions), but not so strong that it breaks the data into brittle micro-clusters. This is consistent with why we see practical separation and interpretable spatial structure in both the  $K = 8$  and  $K = 10$  solutions.

#### 7.1.6 Linear vs Non-Linear Typologies: k-means ( $K = 8$ ) vs Spectral\_nn ( $K = 10$ )

We carried two reference typologies forward:

- **KPCA\_RBF\_g0p001480 + KMeans ( $K = 8$ )** as the main, RESIN-comparable typology (silhouette  $\approx 0.2745$ ).
- **KPCA\_RBF\_g0p001480 + Spectral\_nn ( $K = 10$ )** as a more fine-grained non-linear view (silhouette  $\approx 0.1934$ ).

##### 7.1.6.1 Common structure (what both typologies agree on)

Both methods consistently recover the same macro-patterns:

- a southern dry/heat regime,
- flood-oriented lowlands and riverine corridors,
- a set of high-exposure urban/transport-heavy regions,
- a set of lower-hazard northern/green transition regions.

This agreement is important: it shows the typology is not an artefact of one algorithm.

### 7.1.6.2 Unique structure (what spectral adds)

Spectral clustering mainly adds value by splitting broad groups into more homogeneous subtypes (e.g. separating “urban heat stress” from “urban heavy-precipitation pressure”, or isolating “floodplain expansion” regions). This is visible directly in the spectral typology labels (Table 7.2).

### 7.1.6.3 Why k-means is the main reference

k-means is used as the main reference because it offers the best balance for business use:

- **direct comparability to RESIN** (8 classes),
- **stable partitioning** with understandable cluster narratives,
- **simpler governance** (fewer hyperparameters than spectral graph construction).

Spectral is best positioned as a robustness and “zoom-in” lens, not as the default communication layer.

## 7.1.7 Typology Narratives Supported by SHAP Driver Analysis

A major risk with unsupervised typologies is that they become hard to explain beyond map colourings. We address this by adding an interpretability layer: we train supervised surrogate models to predict cluster membership and compute SHAP values to identify which features drive each cluster.

Compared to classical feature-importance measures, SHAP has two practical advantages in this project: (i) it provides **directional** insight (which features push membership up/down), and (ii) it produces **cluster-specific** explanations rather than one global ranking only. This is critical for turning clusters into defensible narratives.

### 7.1.7.1 Why Logistic Regression is used for SHAP narratives

SHAP explanations are model-dependent: the clarity and stability of driver narratives depend on the surrogate classifier used to approximate the mapping from the HEV feature space to cluster membership. For this reason, we compared multiple surrogate classifiers (linear SVM, Random Forest, and multinomial Logistic Regression) trained to predict the cluster labels from the same preprocessed features.

In this setting, Logistic Regression offered the best balance between predictive fidelity and interpretability. Random Forests can distribute importance across many correlated indicators and interaction splits, which often produces more diffuse and less stable global driver rankings. Linear SVMs can achieve competitive fidelity, but explanation pipelines can be less direct (especially when probability calibration is required), and explanations were more sensitive to scaling and class imbalance. Logistic Regression yielded the most consistent, directionally stable SHAP summaries across reruns, making it the most suitable choice for stakeholder-facing typology narratives.

### 7.1.7.2 KMeans ( $K = 8$ ) driver-based typology table

**Table 7.1** – KMeans ( $K = 8$ ) typology narratives derived from dominant SHAP drivers.

| Cluster | Dominant Drivers (Top SHAP Features)                                                                                                   | Theme                                                        | Typology Label                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|
| 0       | Lower exposure of settlement population, high urban land cover %, employment share, rail exposure, older population share              | Urban Exposure + Socioeconomic Vulnerability                 | Dense Urban Hotspots (Medium Exposure, Medium Adaptation)   |
| 1       | High coastal settlement exposure, medium precipitation, rail exposure, employment share                                                | Climate Hazard + Transport Exposure                          | Climate-Exposed Transport Corridors                         |
| 2       | Change in urban land cover, longer rail length, higher employment share, population exposure to flooding, drought & temperature shifts | High-Activity Industrial Hubs                                | Economic-Infrastructure Clusters under Urban Transformation |
| 3       | Lower urban settlement cover, heat waves, heavy precipitation, extreme weather shifts, rail length, drought hazard                     | Mixed Climate Pressure                                       | Urban-Industrial Nodes under Compound Hazards               |
| 4       | Heat and drought zone, urban settlement exposure, airports, elderly share, migration vulnerability                                     | Socio-Demographic Vulnerability + Air Transport Hubs         | Ageing Urban Regions with Moderate Exposure                 |
| 5       | Extreme flood history, heat waves, high urbanisation                                                                                   | Compound Flood-Heat Interactions                             | Climate-Sensitive Coastal/Lowland Agglomerations            |
| 6       | Landslide-related settlement exposure, water stress, medium airport density                                                            | Educated but Landslide- and Water-Stress-Exposed Cores       | Developed Nodes under Drought & Hazard Pressure             |
| 7       | Settlement exposure, very heavy precipitation, airports, young population decline                                                      | Dynamic Regions under Extreme Precipitation and Fluvial Risk | Expanding Urban Areas Facing Rainfall Risks                 |

### 7.1.7.3 Spectral\_nn ( $K = 10$ ) driver-based typology table

**Table 7.2** – Spectral\_nn ( $K = 10$ ) typology narratives derived from dominant SHAP drivers.

| Cluster | Dominant Drivers (Top SHAP Features)                                                 | Theme                                          | Typology Label                          |
|---------|--------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|
| 0       | Low ports density, high green share, stable climate conditions                       | Low Hazard & High Adaptation                   | Stable Humid Transition Zones           |
| 1       | High flood exposure, rapid land-cover change, economic activity                      | Hydro-Climate Driven Urban Growth              | Urban Floodplain Expansion Zones        |
| 2       | High heat wave frequency, long dry periods, low green cover                          | Thermal Hazard Dominance                       | Urban Heat-Stressed Regions             |
| 3       | High urban land cover, heavy precipitation, transport density                        | Urban-Climate Pressure                         | Metropolitan Areas Under Climate Stress |
| 4       | Ageing population, low exposure, decreasing green urban share                        | Demographic Vulnerability + Limited Adaptation | Aging Low-Exposure Hinterlands          |
| 5       | High migration status                                                                | Medium Hazard Exposure                         | Migration Zone with Rail Connectivity   |
| 6       | Low exposure, high employment, high education share                                  | Socio-Economic Strength, Low Hazard            | Resilient Northern Urban Centers        |
| 7       | Heat exposure, future drought zone, high economic activity                           | Transport & Heat Exposure Corridors            | Industrial and Heat Climate Corridors   |
| 8       | Low exposure, high drought index, fewer wet days                                     | Dry Transitional Zones                         | Arid Southern Transition Regions        |
| 9       | High settlement exposure, increasing heavy precipitation, moderate urban green share | Fluvial Risk with Moderate Adaptation          | Expanding Riverine Urban Settlements    |

These SHAP-supported narratives make the typology defensible: for each type, we can point to a small set of measurable drivers rather than relying on subjective map interpretation alone.

### 7.1.8 Qualitative Crosswalk to RESIN Macro-Types

For communication purposes, it is useful to describe how our types relate to the RESIN macro-typology. We do not claim a perfect one-to-one mapping (the feature sets differ), but at a high level we observe consistent overlap:

**Table 7.3** – Qualitative crosswalk between RESIN macro-types and this thesis’ data-driven typologies.

| RESIN Macro-Type (Examples)             | Main RESIN Narrative                                               | Closest Matches in Our Typologies (Examples)                                                                                                             |
|-----------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Southern lands                          | Heat, drought, wildfire/landslide + lower adaptive capacity        | <i>Arid Southern Transition Regions (Spec-8), Urban Heat-Stressed Regions (Spec-2), and heat/drought-driven KMeans types</i>                             |
| Lowlands and estuaries                  | Coastal + river flooding, high exposure                            | <i>Urban Floodplain Expansion Zones (Spec-1), Expanding Riverine Urban Settlements (Spec-9), Climate-Sensitive Coastal/Lowland Agglomerations (KM-5)</i> |
| North west urban / Inland and urbanised | Dense urban cores; high exposure; mixed hazards                    | <i>Metropolitan Areas Under Climate Stress (Spec-3), Dense Urban Hotspots (KM-4), Urban-Industrial Nodes under Compound Hazards (KM-3)</i>               |
| Northern lands / transition zones       | Lower hazard, greener land cover; generally higher coping capacity | <i>Stable Humid Transition Zones (Spec-4), Resilient Northern Urban Centers (Spec-6)</i>                                                                 |
| Landlocked and elevated                 | Upland hazards (landslides), mixed exposure, localised processes   | <i>Developed Nodes under Drought &amp; Hazard Pressure (KM-6) and upland/localised hazard-driven subtypes</i>                                            |

This crosswalk is useful for stakeholder alignment: it shows that the data-driven pipeline reproduces the same big European “risk geography”, while still introducing additional, investment-relevant refinements (transport corridors, urban growth in floodplains, migration-linked connectivity patterns).

### 7.1.9 LLMs for Cluster Naming and Narrative Labelling

A practical barrier in typology work is that naming clusters typically requires climate domain experts and repeated manual iteration. In this project, SHAP gives structured inputs (top drivers + direction) that can be converted into consistent narratives.

This creates a clear opportunity for **LLM-assisted typology labelling**:

- Provide the LLM with (i) cluster profiles (means/quantiles), (ii) SHAP top drivers, and (iii) constraints such as “avoid value judgments” and “use hazard–exposure–vulnerability language”.
- Ask it to propose short labels and 3–5 sentence narratives per cluster.
- Keep a human-in-the-loop review step for governance and to prevent over-interpretation.

The novelty is not “letting an LLM decide risk”, but using it as a **standardisation tool** to produce consistent, auditable labels when expert time is limited, while keeping the underlying drivers explicit and traceable.

### 7.1.10 Limitations and sources of uncertainty

Several limitations should be considered when interpreting the results. First, the typology is constructed at NUTS3 resolution; spatial aggregation can smooth local extremes and may

hide sub-regional variation that matters for asset-level exposure. Second, indicator engineering relies on proxies and multi-source datasets with heterogeneous coverage, definitions, and measurement error. While missingness handling and robust scaling improve stability, they introduce modelling assumptions that can influence boundary regions and transition zones. Third, the clustering solutions are sensitive to representation choices (e.g. KPCA kernel scale  $\gamma$ ) and, for graph-based methods, to similarity construction; for this reason, stability checks and cross-model comparisons are necessary complements to internal metrics.

Finally, the resulting typology should not be interpreted causally. Clusters reflect recurring patterns in the chosen hazard, exposure, and vulnerability indicators, and are best viewed as a structured *risk segmentation* layer for screening and communication rather than as a deterministic predictor of realised losses.

### 7.1.11 Link Back to the Research Questions

- **RQ1 (Data integration):** the main lesson is that robust preprocessing (coverage checks, outliers, scaling, correlation pruning) is not a side task; it is the core requirement for stable typologies.
- **RQ2 (Risk scoring):** composite risk scores and clusters are complementary: scores rank regions, clusters explain *why* regions are similar (heat/drought vs flood/exposure vs vulnerability).
- **RQ3 (Typology construction):** the selected KPCA + k-means ( $K = 8$ ) solution provides an interpretable, RESIN-comparable typology; spectral clustering confirms the macro-structure and adds fine-grained subtypes.
- **RQ4 (Robustness):** failure modes (e.g. DBSCAN degeneracy, “too-good-to-be-true” silhouette in PCA) are informative; robust typology selection must combine metrics with cluster-size sanity checks and spatial coherence.
- **RQ5 (Business integration):** SHAP-supported narratives and a RESIN crosswalk make the typology usable as a decision-support layer (screening + explanation), not just a research output.

### 7.1.12 Implications for investment screening and governance

For investment workflows, the typology is most valuable as a decision-support layer. At portfolio level, it enables risk concentration analysis (how much exposure sits in flood-dominated vs. heat/drought-dominated regimes). At asset level, it supports triage: regions in high-exposure or high-vulnerability clusters can be flagged for deeper due diligence (e.g. flood protection, drainage capacity, building-level adaptation measures, insurance terms, and local resilience planning). Governance-wise, the combination of (i) transparent indicators, (ii) stability checks, and (iii) SHAP-backed narratives provides an auditable basis for periodic re-runs as new data, updated projections, or scenario assumptions become available.

Overall, the evidence supports the central claim of the thesis: a carefully engineered hazard–exposure–vulnerability feature space, combined with non-linear representation learning (KPCA) and robust clustering, can reproduce the main expert typology patterns while adding operational detail that is directly relevant for real-estate risk decisions.

# Chapter 8

## Conclusion

### 8.1 Conclusion

This thesis set out to build a scalable, data-driven framework for assessing climate risk at the regional level in Europe and to test whether unsupervised learning can replicate or challenge existing expert-defined climate typologies. The work combined large-scale climate and socio-economic data, a structured hazard–exposure–vulnerability framework, an extensive preprocessing pipeline, and a set of clustering and validation methods.

This chapter summarises the main contributions, reflects on what has been learned, and outlines how the approach can be used beyond the original climate typology task.

#### 8.1.1 Main Contributions

##### 8.1.1.1 A Complete Hazard–Exposure–Vulnerability Dataset

The first contribution is the construction of a comprehensive climate risk dataset at NUTS3 level for most of Europe. Using openly available sources, the project brought together:

- climate hazard indicators (temperature trends, heat and drought indices, heavy precipitation, river flooding, coastal surge, wildfire and landslide proxies),
- exposure indicators (population, built-up area, residential footprint, population and buildings in hazard-prone zones), and
- vulnerability indicators (age structure, income and labour-market proxies, access and infrastructure-related variables).

A large share of the work went into cleaning, harmonising and scaling these indicators so that they could be used together in a single analytical pipeline. Outlier treatment, robust scaling and correlation checks were crucial to avoid spurious structure. The result is a consistent feature matrix that captures both physical and socio-economic dimensions of climate risk and can be re-used for future studies.

##### 8.1.1.2 Dual Algorithmic Typology: Linear and Non-Linear Views

The second contribution is the development of an algorithmic climate risk typology based on this integrated dataset. By applying dimensionality reduction and clustering techniques to the hazard–exposure–vulnerability space, the project produced two complementary regional climate risk typologies:

- a **linear** k-means solution with  $K = 8$  clusters in PCA space, used as the main reference typology; and

- a **non-linear** spectral clustering solution with  $K = 10$  clusters in kernel PCA space, used as a finer-grained view.

The k-means  $K = 8$  typology offers very strong internal compactness and separation in the PCA space and yields eight clusters that map directly to clear climate risk narratives (“hot and dry southern regions”, “coastal and river flood belts”, “highly exposed but wealthy metros”, “peripheral high-vulnerability regions”, “mountain and upland regions”, etc.). The spectral  $K = 10$  typology preserves the same macro-structure but splits several clusters into more homogeneous subtypes, revealing additional non-linear structure where the data support it.

Together, these two solutions show that regional climate risk patterns can be described at different levels of granularity using a single, consistent pipeline.

### 8.1.1.3 External Validation, Robustness and Spatial Coherence

A third contribution is the careful validation of the clustering results, both internally and against the expert typology.

Internally, we used silhouette, Calinski–Harabasz and Davies–Bouldin indices to screen out degenerate and overfitted solutions. Very small or fragmented clusters were discarded even when their scores were superficially high. The final k-means and spectral solutions achieve a good balance between internal quality and interpretability.

Externally, we compared all candidate solutions to the existing expert-based European climate typology using Adjusted Rand Index and Normalised Mutual Information. The k-means and spectral reference models show clearly positive ARI and NMI values, well above random expectation, even though the expert labels were not used during training and the indicator sets are not identical. This confirms that the data-driven typologies recover a meaningful share of the expert-defined structure.

Mapping the clusters across Europe reveals spatially coherent regions that line up with known climate regimes and geographical features. Neighbouring NUTS3 units are typically assigned to the same or related types, which supports the idea that the typology is capturing real large-scale climate and socio-economic patterns rather than noise. Sensitivity experiments with alternative scaling, feature subsets and algorithms show that the macro patterns are robust.

### 8.1.1.4 Generalisation to Other Scenarios and Domains

Although the main focus of the thesis is a pan-European climate risk typology, the underlying methodology has also been tested on other questions, such as:

- scenario-based analysis of flood-related risk, where the same pipeline is applied with a stronger emphasis on river and coastal indicators, and
- a housing affordability study for German cities, where the clustering logic is reused with indicators describing rents, prices, incomes, migration and housing stock characteristics.

These applications show that the core ideas of the pipeline — careful data engineering, dimensionality reduction, clustering and indicator-based interpretation — are not tied to a single topic. They can be adapted to different domains as long as there is a clearly defined feature space and a meaningful way to interpret groups. In that sense, the project delivers a generic “multi-lens” clustering engine that can be pointed at climate, affordability or other structural questions.

### 8.1.1.5 An Interpretable and Reproducible Decision-Support Tool

Finally, the project links the technical work back to real-world use by integrating the regional risk scores and clusters into an interactive dashboard for Catella. Users can:

- explore the climate risk landscape at European and country level,
- inspect the risk type and composite risk score for individual regions and assets, and
- compare how different assumptions or indicator weights affect the classification.

This turns a complex modelling pipeline into a practical decision-support tool. The approach is transparent and reproducible: all steps, from raw data to final cluster label, are documented and can be re-run when new data, new scenarios or new investment questions arise.

### 8.1.2 Overall Assessment

Taken together, the results demonstrate that unsupervised learning is a viable way to construct a climate risk typology that is consistent with expert judgement but also adds new insights.

The algorithm recovers the main patterns that climate experts expect to see, yet it is more systematic in how it combines hazards with exposure and vulnerability. The explicit comparison between a linear k-means typology and a non-linear spectral typology shows that the core message is robust to different modelling choices: Mediterranean heat and drought, major river and coastal flood belts, highly exposed metropolitan regions and socio-economically fragile peripheral regions emerge in all reasonable configurations.

At the same time, the work makes clear that very high internal validation scores can be misleading when they are driven by tiny or unstable clusters. The final model selection is based not only on metrics, but also on stability, spatial coherence and interpretability. This is an important lesson for any data-driven risk classification: numerical scores are useful, but they are not sufficient on their own.

The thesis also underlines that the success of this kind of approach depends heavily on data engineering. Without robust preprocessing, outlier treatment and sensible feature selection, clustering algorithms are unstable and hard to interpret. With a clean and well-structured feature space, they become powerful tools for summarising complex risk landscapes and for challenging or updating expert views.

### 8.1.3 Answers to the Research Questions

**RQ1 (Data integration and representation):** Addressed by constructing a unified NUTS3-level feature matrix that integrates heterogeneous hazard, exposure, and vulnerability indicators, supported by consistent spatial aggregation, harmonisation, and robust preprocessing.

**RQ2 (Climate risk scoring):** Addressed through the development of a composite regional risk scoring layer that translates multi-dimensional HEV indicators into comparable summary measures, enabling portfolio screening and asset-linked interpretation alongside typology membership.

**RQ3 (Algorithmic typology construction):** Addressed by systematically testing multiple embedding-clustering configurations and selecting stable reference solutions, demonstrating measurable agreement with expert typologies and recovery of coherent macro-scale spatial structure.

**RQ4 (Interpretability and scenario extensions):** Addressed by translating clusters into defensible narratives using indicator-based profiles and a SHAP explanation layer via a surrogate classifier. Among tested surrogates (SVM, Random Forest, Logistic Regression), Logistic Regression provided the clearest and most stable global driver summaries. The same pipeline

was also shown to generalise to additional scenario studies, including a flood-focused analysis and an affordability typology.

**RQ5 (Business integration and dashboard design):** Addressed by deploying risk scores and typology outputs in an interactive dashboard that supports both top-down regional screening and bottom-up asset checks, including sensitivity to modelling assumptions and indicator weights.

#### 8.1.4 Outlook

This thesis does not claim to provide the final word on regional climate risk in Europe. Instead, it offers a framework that can evolve.

As new climate projections, improved hazard models and richer socio-economic data become available, they can be fed into the same pipeline. Additional hazards such as pluvial flooding, more detailed heatwave metrics or more granular climate model ensembles can be added as extra indicators. Likewise, future work can extend the approach to dynamic settings, for example by clustering time series of indicators to track how regions move between types over time.

On the application side, the same logic can be used for more targeted scenario studies (for example, flood-heavy stress tests) or for combined “multi-lens” analyses that integrate climate risk, affordability and other structural factors into a single view. At asset level, linking the regional clusters and composite scores to building-level characteristics opens up a path towards more detailed adaptation planning.

The key message is that combining a hazard–exposure–vulnerability perspective with transparent, data-driven methods creates a bridge between climate science and financial decision-making. For an asset manager, this means that climate risk is no longer a vague background concern but a structured, mappable and explainable dimension of regional and asset selection.

In that sense, the main contribution of this work is not only the specific typology produced for this project, but also the demonstration that algorithmic climate risk classification can be made robust, interpretable and directly usable in real-estate investment practice, and that it can be extended to new datasets, new scenarios and new questions without redesigning the entire approach from scratch.

# Chapter 9

## Future Work

### 9.1 Future Work

This thesis shows that a data-driven hazard–exposure–vulnerability framework can produce a meaningful climate risk typology for European regions and can be linked to real-estate investment questions. At the same time, several simplifications were necessary to keep the project manageable. This section outlines concrete directions for future work that could extend and strengthen the approach. In particular, the future work directions below directly address the main sources of uncertainty identified in this study: spatial aggregation effects at NUTS3 level, proxy-based exposure and vulnerability indicators, missing-data assumptions, and climate-model uncertainty (single-model projections). Strengthening these aspects would increase both scientific robustness and business usability of the typology.

#### 9.1.1 Richer Hazard Representation

The current indicator set focuses on a core group of hazards: temperature, precipitation, river flooding, coastal flooding, drought, wildfire and landslide potential. Several additional dimensions could be added to better capture local risk, especially in dense urban areas.

##### 9.1.1.1 Pluvial Flooding and Urban Drainage

At present, flood indicators mainly represent fluvial (river) flooding and coastal surge. Intense local rainfall and pluvial flooding in urban environments are only indirectly captured. Future work could integrate:

- high-resolution pluvial flood maps, where available, and
- urban drainage and impermeable-surface information derived from land cover and remote sensing.

This would be particularly relevant for metropolitan regions where local drainage capacity and surface sealing are major drivers of loss.

##### 9.1.1.2 More Detailed Heat and Humidity Indicators

Heat risk is currently represented through standard indices such as summer days and heat waves. Additional indicators could capture:

- combined heat and humidity stress (e.g. apparent temperature or wet-bulb temperature),
- night-time minimum temperatures during heat events, which are critical for health impacts, and
- urban heat island effects derived from remote-sensing surface temperature products.

Such refinements would help distinguish between regions with similar mean warming but very different levels of human heat stress, and would also improve the interpretation of “hot and dry” versus “hot and humid” cluster types.

### 9.1.1.3 More Granular CORDEX/CMIP Deltas and Compound Events

The climate projections used here are based on a single model and scenario at daily resolution. Future work could:

- combine multiple regional or global models to construct multi-model ensembles and quantify spread,
- use more granular deltas across time slices (e.g. 2030s, 2050s, 2080s) to analyse how typologies evolve under different warming levels, and
- add indicators for compound events (e.g. heat + drought, storm + coastal surge) where joint occurrence is important for loss.

This would strengthen the treatment of physical uncertainty and support more explicit scenario analysis.

## 9.1.2 Advanced Dimensionality Reduction and Representation Learning

This thesis used Principal Component Analysis as the main dimensionality reduction method, with kernel PCA used for the spectral clustering variant. PCA has the advantage of simplicity and interpretability, but it is strictly linear.

Future work could experiment with:

- non-linear manifold learning techniques such as diffusion maps or UMAP variants tuned for clustering stability,
- autoencoders and other neural representation-learning models to capture more complex interactions between indicators, and
- hybrid approaches where a non-linear method is used to explore structure, and a simpler linear model is used for explainability and for mapping back to indicators.

Any new representation should still allow clear links back to original indicators and risk drivers. A useful benchmark would be whether alternative embeddings can improve external validation or spatial coherence compared to the current PCA / kernel PCA setup.

### 9.1.3 Improved Model Selection and Consensus Typologies

The final typology in this thesis is based on k-means with  $K = 8$  clusters in PCA space, with spectral clustering ( $K = 10$  in kernel PCA) used as a more detailed alternative. Model selection relied on a combination of internal indices, external validation against the expert typology and visual inspection.

Future work could make this step more systematic by:

- using stability-based criteria (e.g. resampling or perturbation tests) to quantify how sensitive each configuration is to noise and preprocessing choices,
- applying consensus clustering to combine several high-performing solutions into a single, more stable consensus typology, and
- explicitly reporting confidence measures for each region (for example, how often it changes cluster across near-equivalent solutions).

Such tools would help communicate uncertainty in the typology itself, not only in the underlying climate indicators.

### 9.1.4 Validation Against Observed Impacts and Financial Loss Data

A major next step is to link the typology to observed outcomes. While this thesis validates structure against expert typologies and clustering metrics, investor relevance is ultimately determined by realised impacts. Future work could:

- link regional types and scores to historical loss proxies (e.g. flood damage estimates, insurance claims where accessible, disaster databases, or public loss reports),
- test whether high-risk clusters correspond to higher frequency or severity of observed events and losses, and
- calibrate simple damage functions (expected loss curves) per hazard regime to move from *classification* towards *impact quantification*.

This would also help translate typologies into financial metrics (e.g. expected annual loss or stress-test losses) that are directly comparable across regions and assets.

### 9.1.5 Multi-Lens Clustering: Climate, Affordability and Beyond

The methodology developed here is not limited to climate risk. The parallel work on housing affordability and the flood scenario study already shows that the pipeline can be reused with different indicator blocks.

A natural next step is to build a joint “multi-lens” clustering framework that integrates both climate indicators and affordability or market indicators at city or regional scale. This would allow:

- simultaneous segmentation of regions by climate risk and housing-market stress, and
- identification of hotspots where high physical risk overlaps with tight housing markets or vulnerable tenant populations.

Technically, this would involve:

- extending the feature set with affordability variables (prices, rents, incomes, construction, vacancy, net migration),
- testing different weighting schemes between the climate and affordability blocks, and
- exploring multi-view or multi-block clustering methods that can handle several related but distinct indicator groups.

The same idea could be extended further to include transition risk, regulatory indicators or energy performance, so that clusters reflect multiple dimensions of risk that matter for real-estate portfolios.

### 9.1.6 Temporal Models and Dynamic Clustering

The current typology is essentially static: indicators are aggregated over a future time window and used to assign a single type per region. In reality, both climate hazards and socioeconomic conditions change over time.

Future work could explore dynamic approaches, for example:

- constructing time series of indicators (historical and projected) and applying clustering to trajectories rather than single snapshots,
- using temporal clustering or sequence clustering methods to identify typical evolution paths of regions (e.g. “transitioning from moderate to high heat stress”), and
- linking these trajectories to adaptation policy timelines and investment horizons.

This would shift the focus from “What type is this region now?” to “Where is this region heading under different scenarios, and how fast?”

### 9.1.7 Cluster Evolution under SSP Pathways

Related to temporal dynamics, the framework could be extended to explicitly compare cluster structures under different Shared Socioeconomic Pathways (SSPs) and emissions scenarios.

For example, future work could:

- re-run the pipeline under multiple SSP–RCP combinations,
- track how many regions move from one cluster type to another as the climate warms, and
- quantify how mitigation or adaptation assumptions in the SSPs change the distribution of high-risk types and the location of hotspots.

This would provide a more policy-relevant view of climate risk, connecting regional typologies to global scenario choices and to the timing of expected changes.

### 9.1.8 Automated Monitoring and Dashboard Extensions

Finally, there is substantial potential to turn the current one-off analysis into an ongoing monitoring system.

Future work could:

- automate data updates and re-runs of the full pipeline when new climate data, socio-economic statistics or asset information become available,
- extend the dashboard with time sliders, scenario selectors and alert mechanisms (e.g. highlighting regions whose risk type or composite score has changed since the last run), and
- integrate asset-level adaptation measures, cost estimates and forward-looking metrics (e.g. expected damage under selected scenarios) to move from risk classification towards concrete action planning.

This would transform the typology from a static research product into a living tool that supports continuous risk management and strategic decision-making.

In summary, the future work outlined above demonstrates that the current thesis serves as a strong baseline rather than a final endpoint. By adding richer hazards, exploring more advanced representations, improving model selection, combining climate with affordability, and extending the framework in time, the same basic methodology can support an increasingly detailed and forward-looking view of climate risk for European real estate.

# Appendix A

## Appendix

### A.1 Clustering Experiment Overview

This section provides a representative subset of clustering experiments executed during development. It demonstrates breadth across algorithm families (centroid-based, hierarchical, model-based, graph-based, and density-based), across embeddings, and across plausible cluster counts. The objective is transparency: showing that the final configuration was selected from a wider and systematically evaluated search space.

*Note:* Silh. denotes the Silhouette coefficient (higher is better), CH the Calinski–Harabasz index (higher is better), and DB the Davies–Bouldin index (lower is better). Trust refers to embedding trustworthiness where applicable.

**Table A.1** – Representative clustering experiments (core metrics).

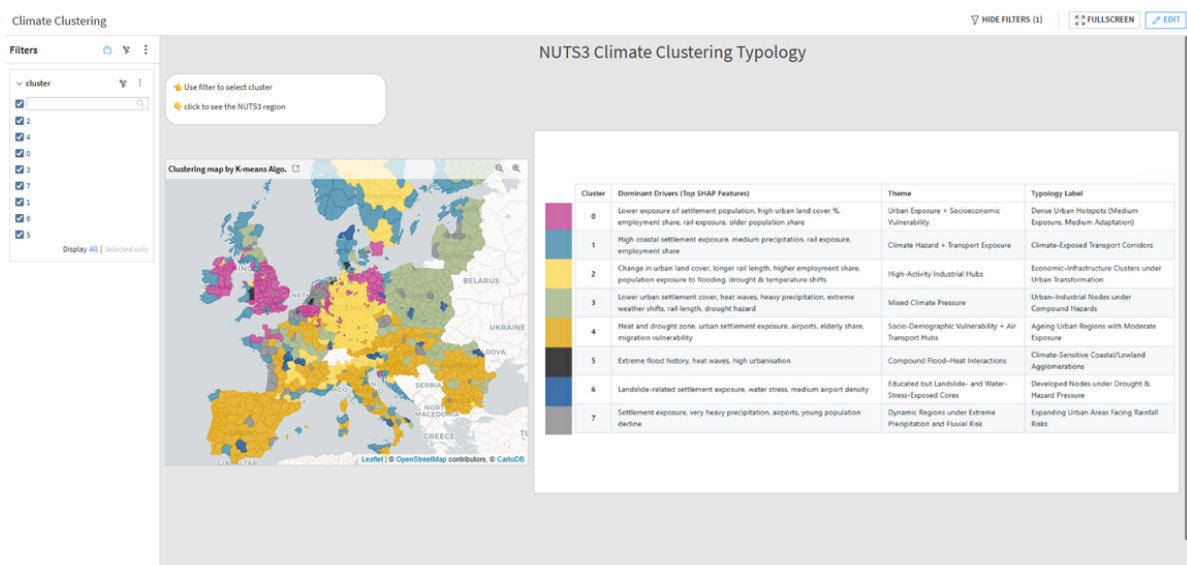
| Embedding         | Stage | Algorithm                       | Family | $k$ | $N_C$ | Silh. | CH     | DB    | Trust |
|-------------------|-------|---------------------------------|--------|-----|-------|-------|--------|-------|-------|
| PCA-k14           | AGG   | Agglomerative (average linkage) | Hier.  | 8   | 8     | 0.978 | 131391 | 0.409 | 0.572 |
| PCA-k14           | AGG   | Agglomerative (average linkage) | Hier.  | 10  | 10    | 0.979 | 227922 | 0.397 | 0.572 |
| PCA-k14           | AGG   | Agglomerative (Ward linkage)    | Hier.  | 8   | 8     | 0.979 | 153473 | 0.405 | 0.572 |
| PCA-k14           | KM    | KMeans                          | Cent.  | 10  | 10    | 0.980 | 232503 | 0.394 | 0.572 |
| RAW-full          | GMM   | Gaussian Mixture (diag cov.)    | Model  | 6   | 6     | 0.911 | 16289  | 2.560 | –     |
| KPCA-RBF-0.00148  | DBS   | DBSCAN                          | Dens.  | –   | 2     | 0.836 | 5136   | 0.132 | 0.998 |
| KPCA-RBF-0.004933 | OPT   | OPTICS                          | Dens.  | –   | 3     | 0.898 | 2132   | 0.335 | 0.998 |
| KPCA-RBF-0.00148  | SPEC  | Spectral NN                     | Graph  | 2   | 2     | 0.772 | 2321   | 0.322 | 0.998 |

## A.2 Climate Dashboards

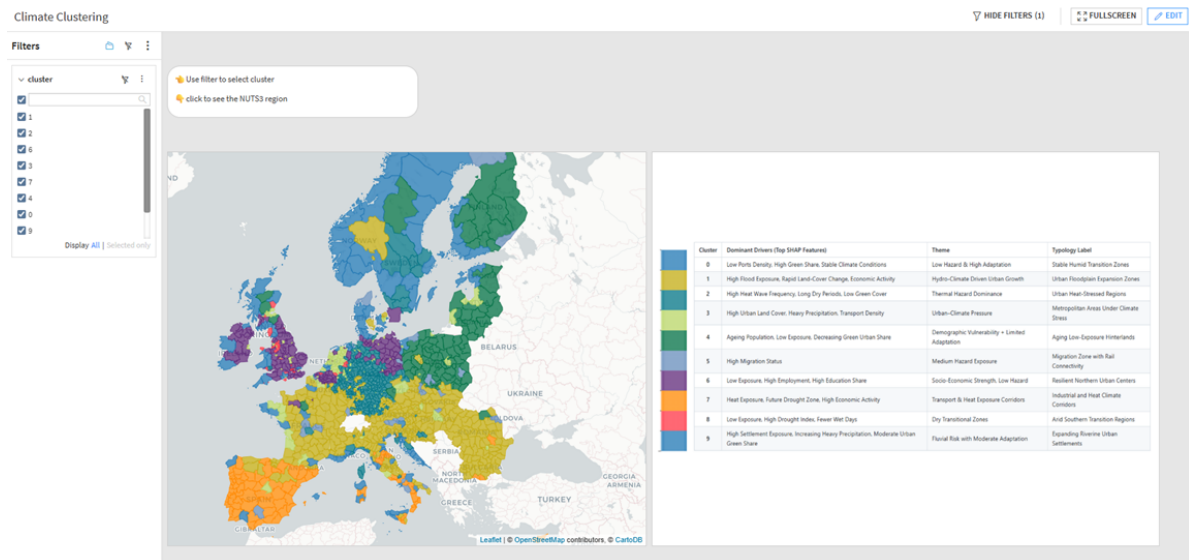
This section documents the climate dashboards developed during the project. Screenshots are included for reproducibility and to communicate the end-user workflow: from region-level typologies to scenario overlays and asset-level drill-down.

**Novelty note (project-level).** Beyond producing a static typology, the work operationalises a full ML-to-dashboard pipeline: (i) clustering outputs are directly consumable by stakeholders, (ii) scenario overlays (flood-focused) reuse the same regional framework to support “what-if” exploration, and (iii) the dashboard design enables interpretation without requiring climate-domain expertise at every step (e.g., consistent cluster labels, contextual tables, and guided navigation).

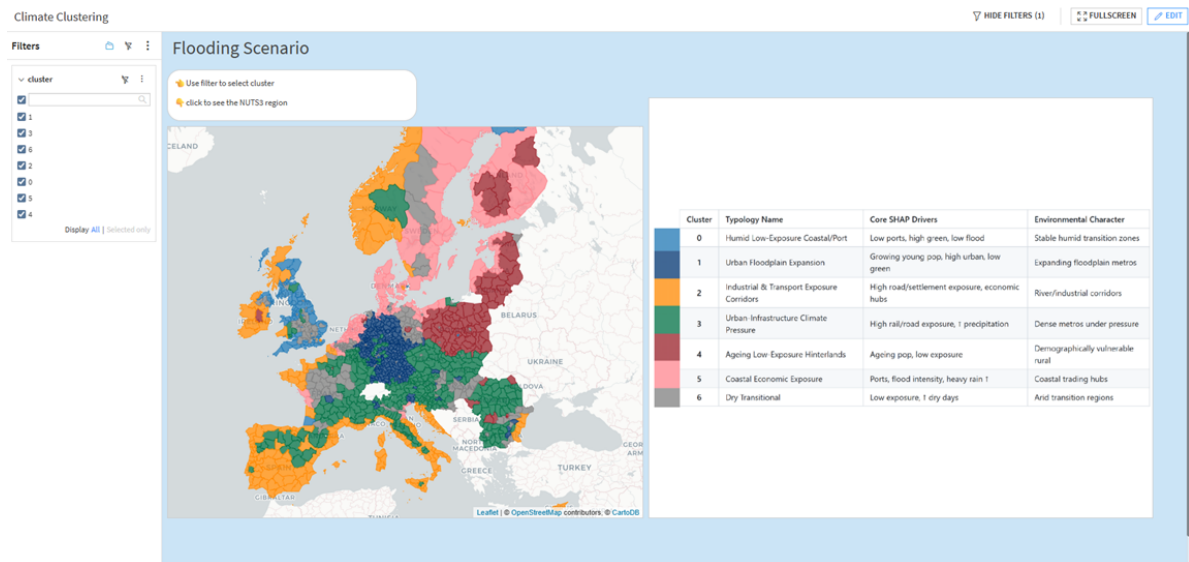
### A.2.1 Climate Typology and Flooding Scenario Dashboard



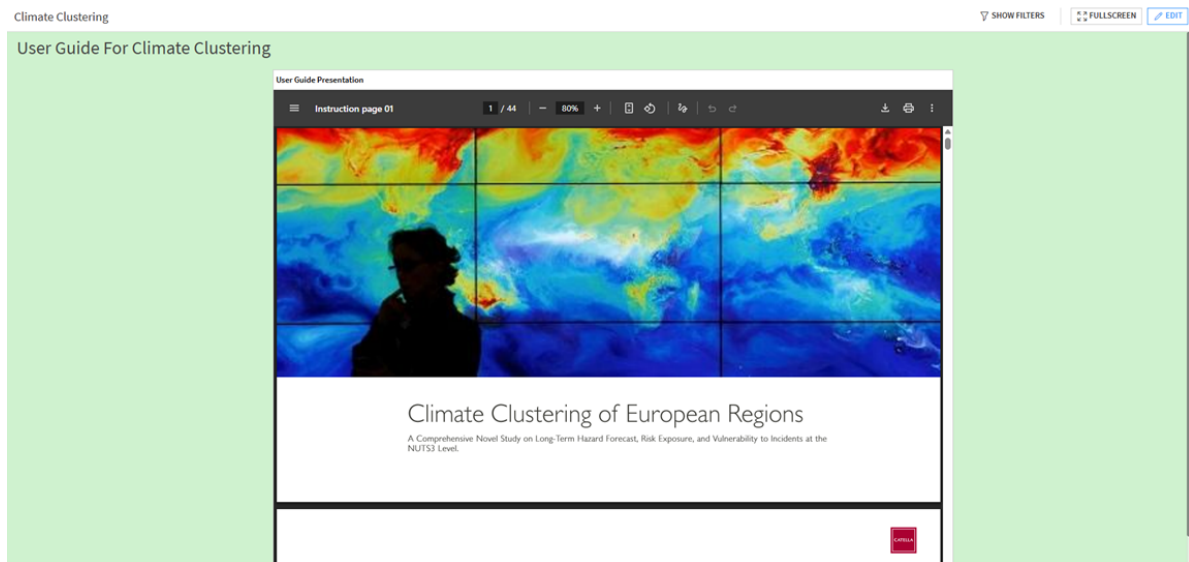
**Figure A.1** – Climate clustering dashboard: NUTS3 typology map with cluster legend and label table.



**Figure A.2** – Climate clustering dashboard: alternative typology view highlighting spatial continuity.



**Figure A.3** – Flooding scenario dashboard: flood-focused typology overlay using the same regional framework.



**Figure A.4** – User guide cover page for the climate clustering dashboard.

## A.2.2 Asset Climate Dashboard (CRIM Assets)

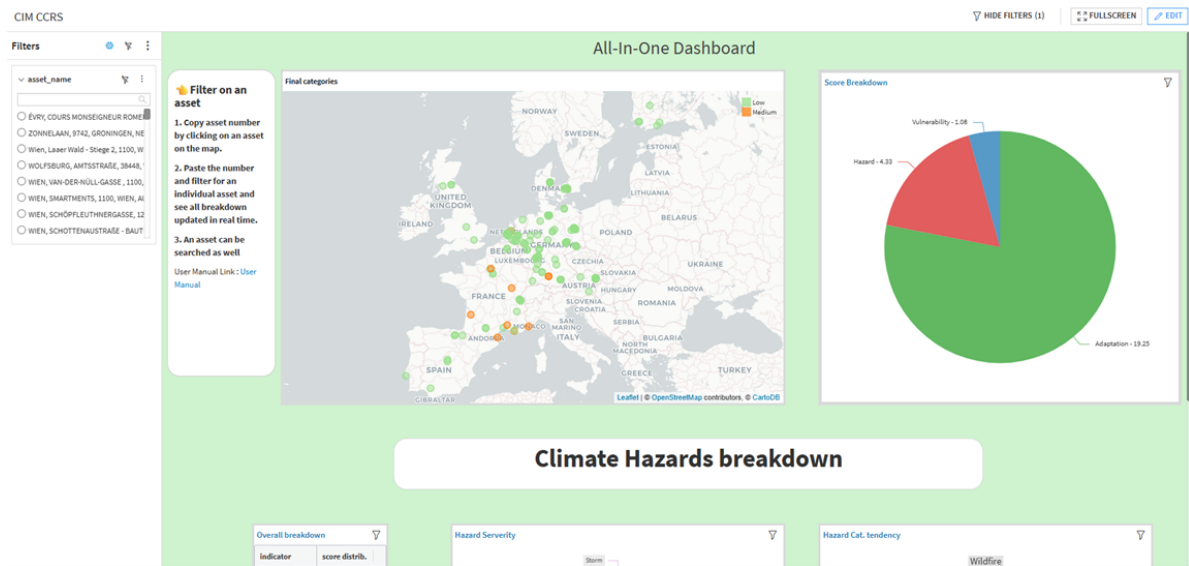


Figure A.5 – All-in-one dashboard: portfolio map view with filters and high-level overview.

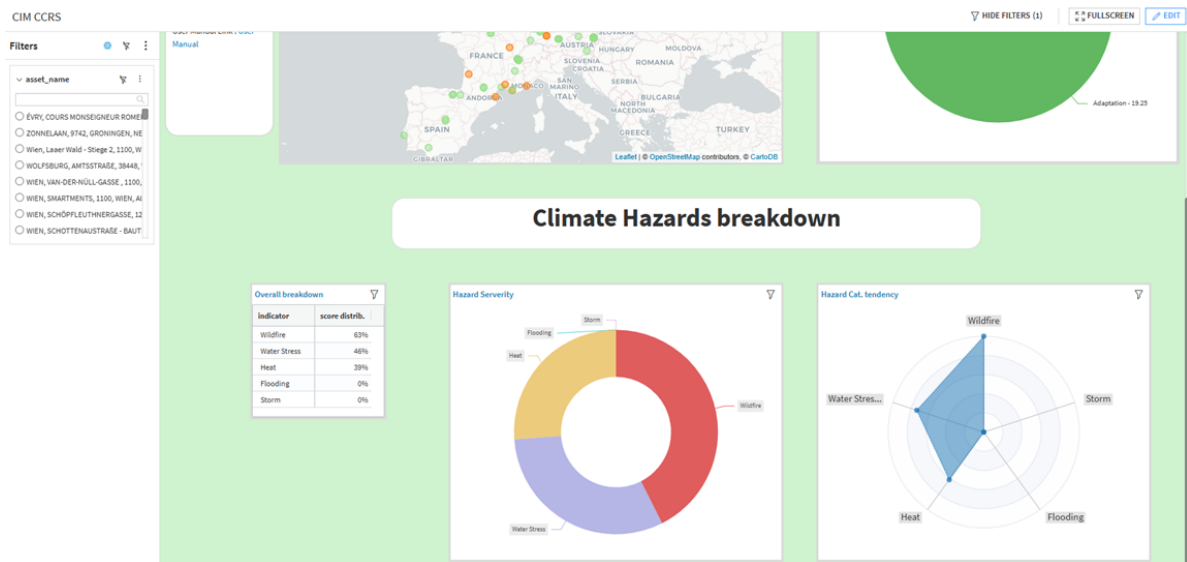


Figure A.6 – Portfolio hazard breakdown: summary views supporting top-down screening.

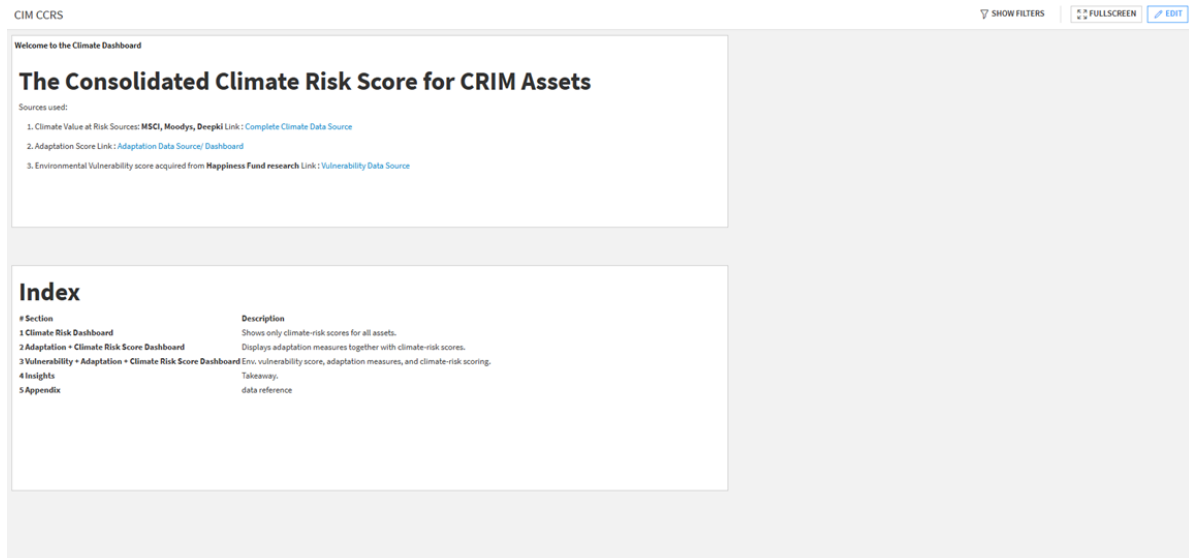


Figure A.7 – Consolidated climate risk score report landing page (index / navigation).

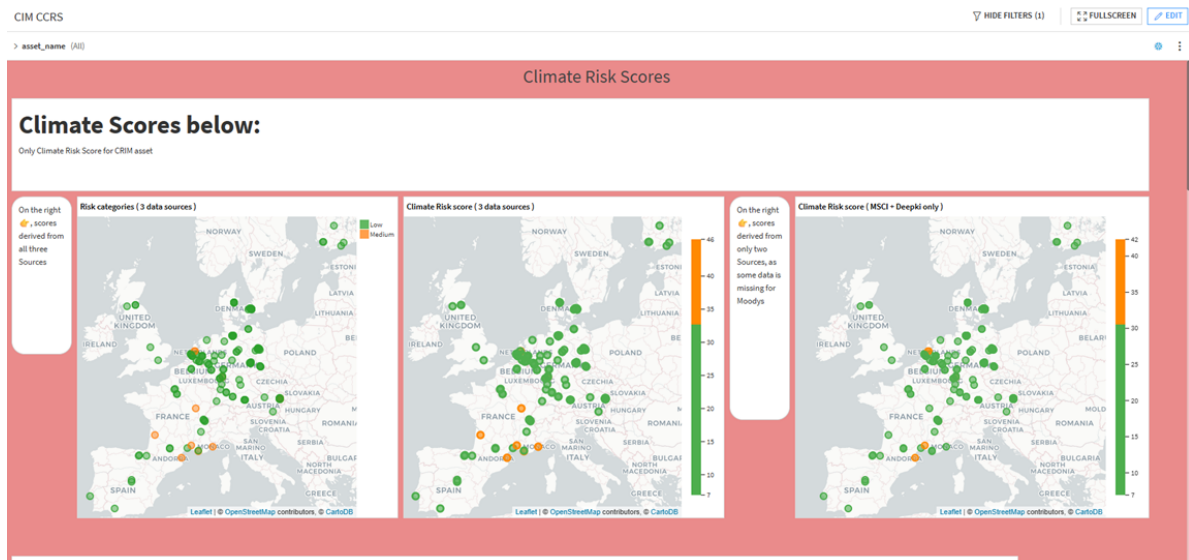


Figure A.8 – Climate risk scores: hazard-specific panels for asset-level comparison.

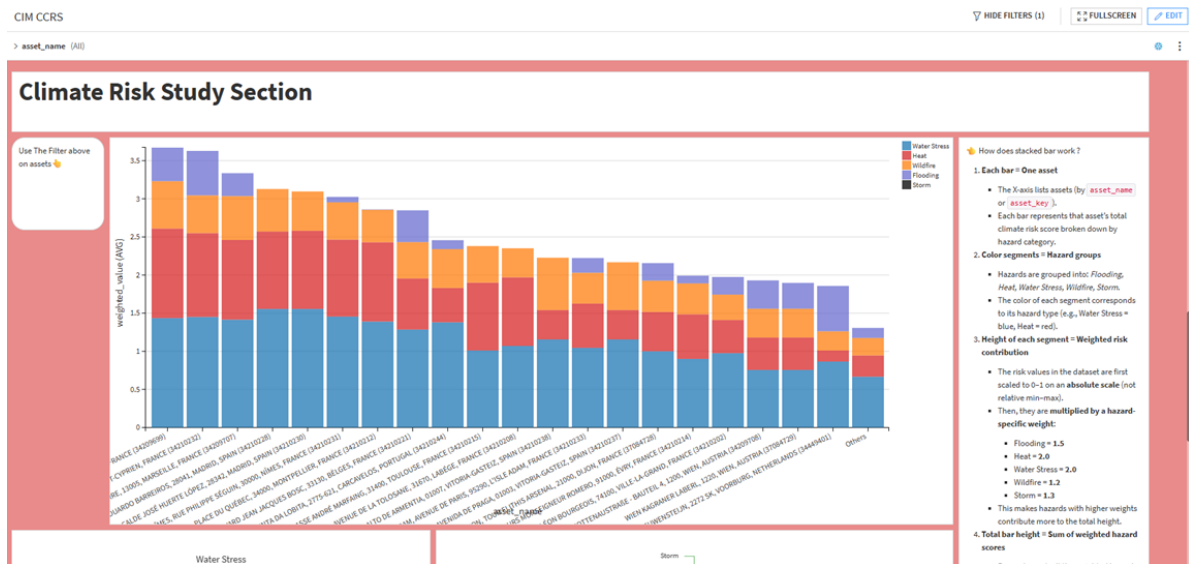


Figure A.9 – Climate risk study section: portfolio ranking/stacked views with filters for drill-down.

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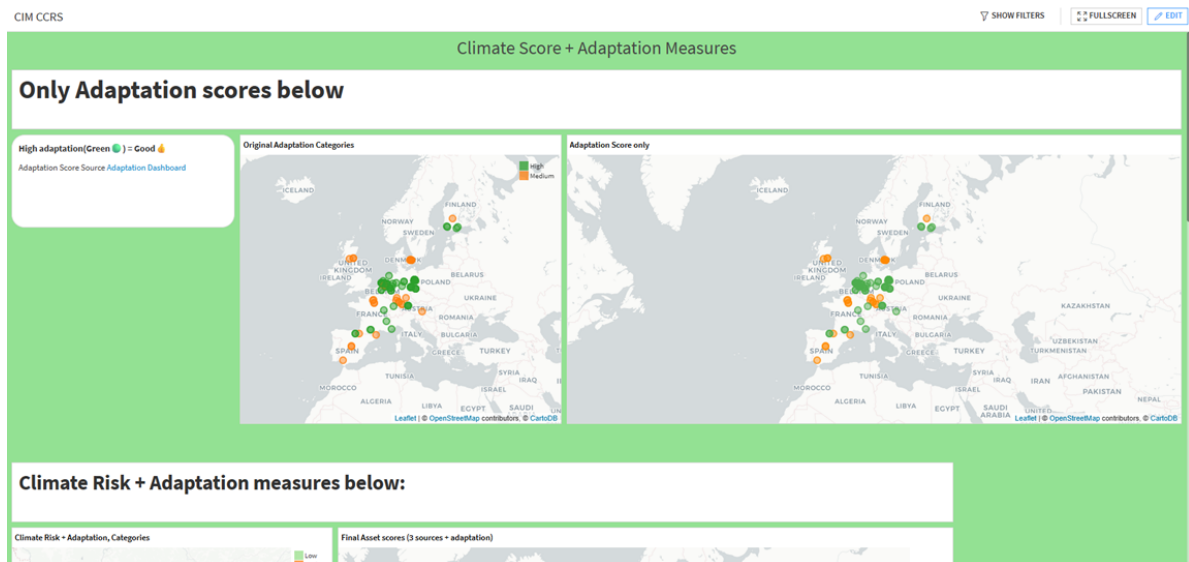


Figure A.11 – Adaptation measures: dashboard view focusing on adaptation scores.

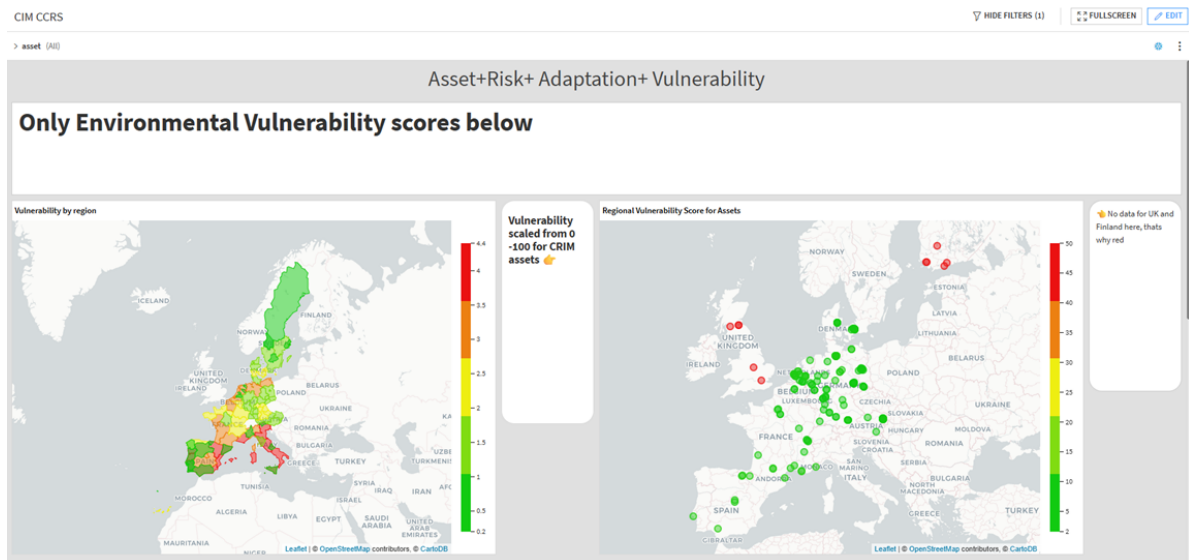
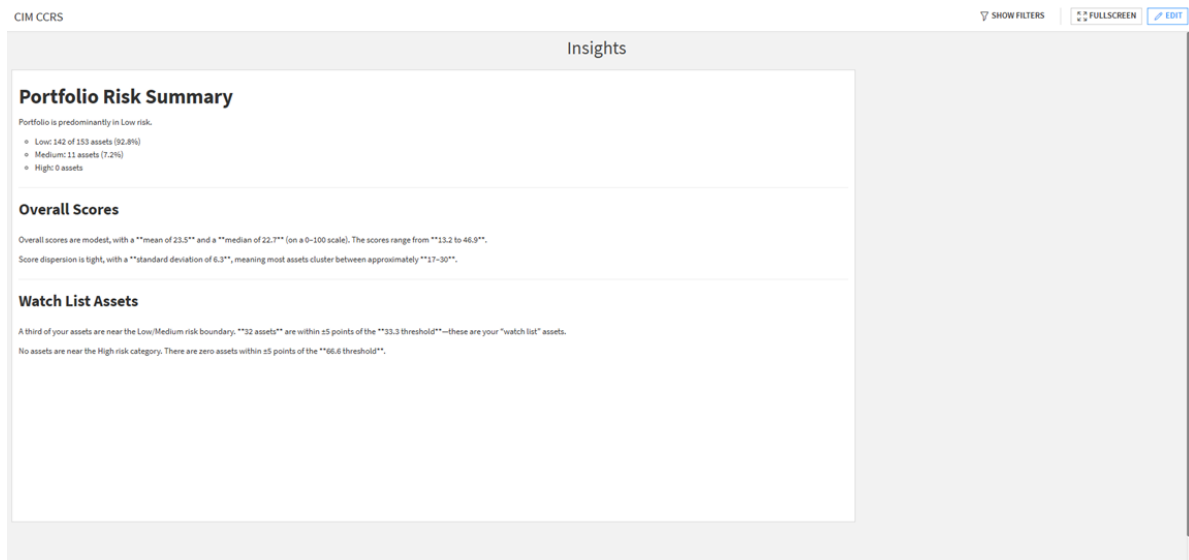


Figure A.12 – Combined view: climate risk + adaptation measures (resilience-oriented interpretation).



**Figure A.13** – Insights: portfolio risk summary, overall scores, and watchlist-style outputs.

### A.3 Affordability Dashboard (Germany)

This dashboard is a separate research stream that applies the same “typology” philosophy to housing affordability: build a consistent city-year panel, transform and scale features robustly, cluster cities into interpretable regimes, and then analyse transitions over time.

**What was done (summary).** A city-level panel (2010–2024/25) was built by integrating city indicators (demographics, stock, construction, tenure), macroeconomic signals, and market indicators (prices/rent/transactions, where available). Skew was reduced via Yeo–Johnson transformation (supports zero/negative values), outlier impact was limited via robust scaling (median/IQR), dimensionality reduction supported stability and visualization, and the final clustering was selected based on internal validity plus interpretability. Transition heatmaps were produced to communicate persistence and regime shifts.

**Novelty note (cross-domain transfer).** The affordability work demonstrates that the pipeline is not “one-off” for climate hazards: the same end-to-end unsupervised typology + transition analysis approach generalises to a different domain with different data-generating processes. This strengthens the methodological contribution: a reusable clustering-and-interpretation framework that can support investor-facing decision workflows across themes (risk, resilience, and market structure).

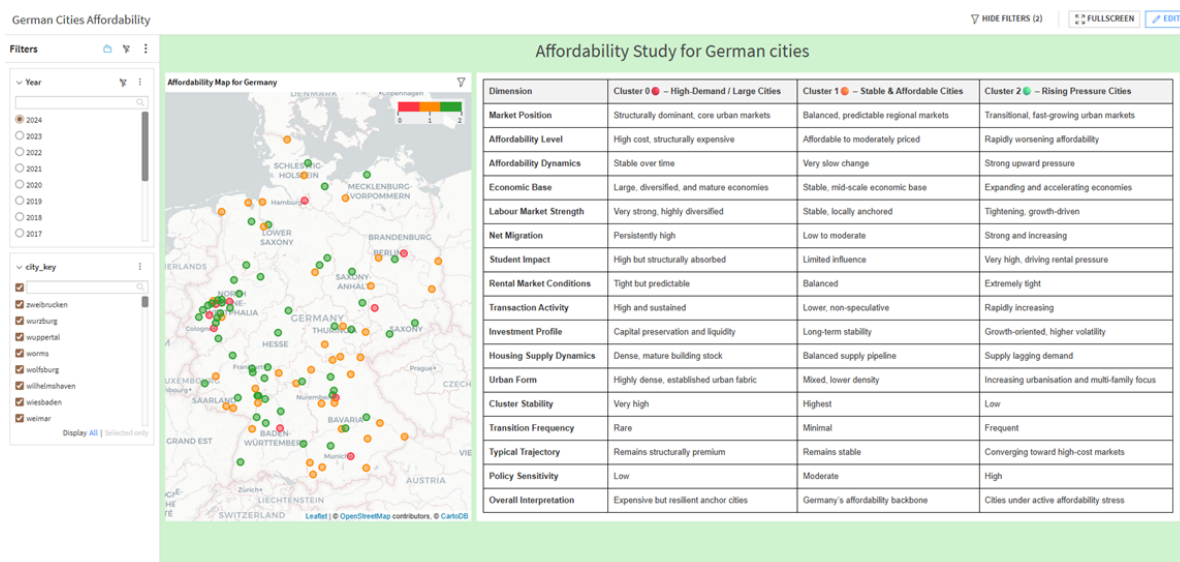
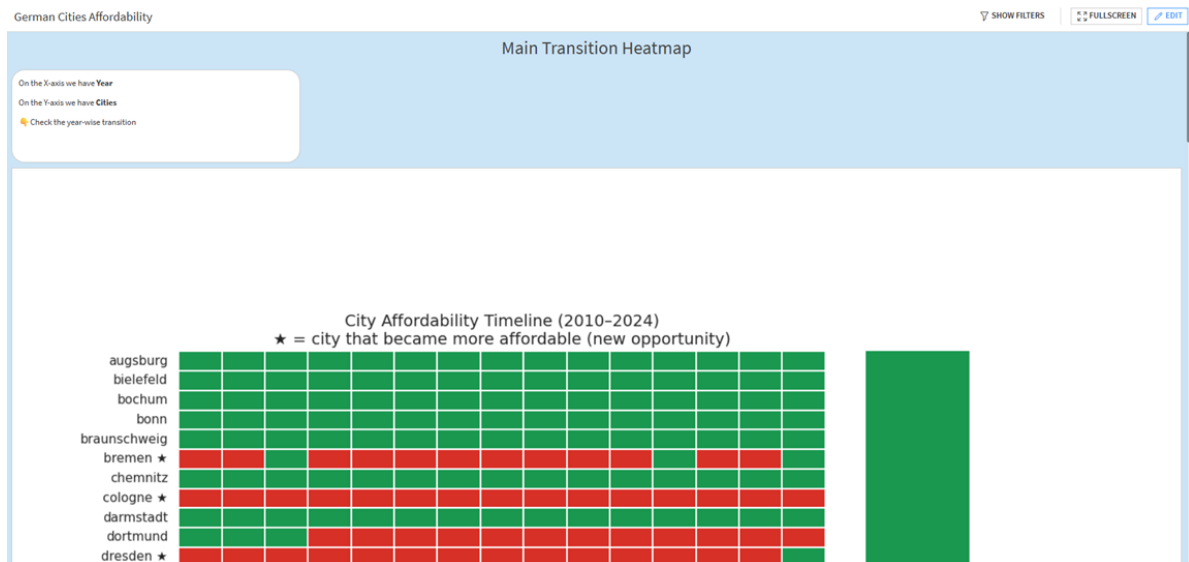


Figure A.14 – Affordability study: dashboard overview with cluster definitions and typology table.



**Figure A.15** – Main transition heatmap (part 1): city affordability timeline (2010–2024).



**Figure A.16** – Main transition heatmap (part 2): continuation to cover the full panel.

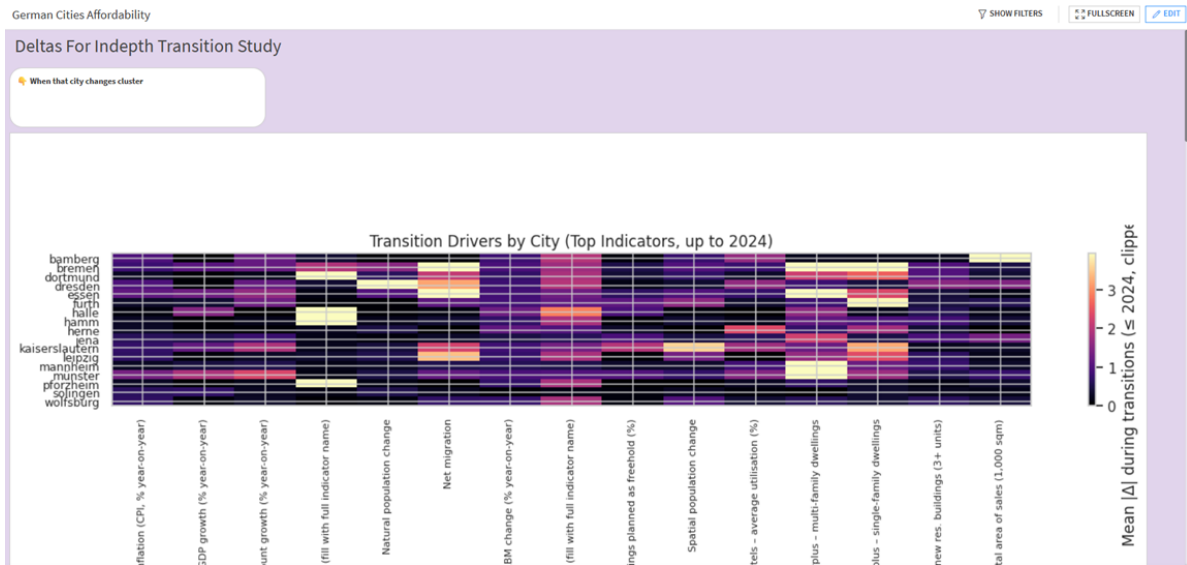
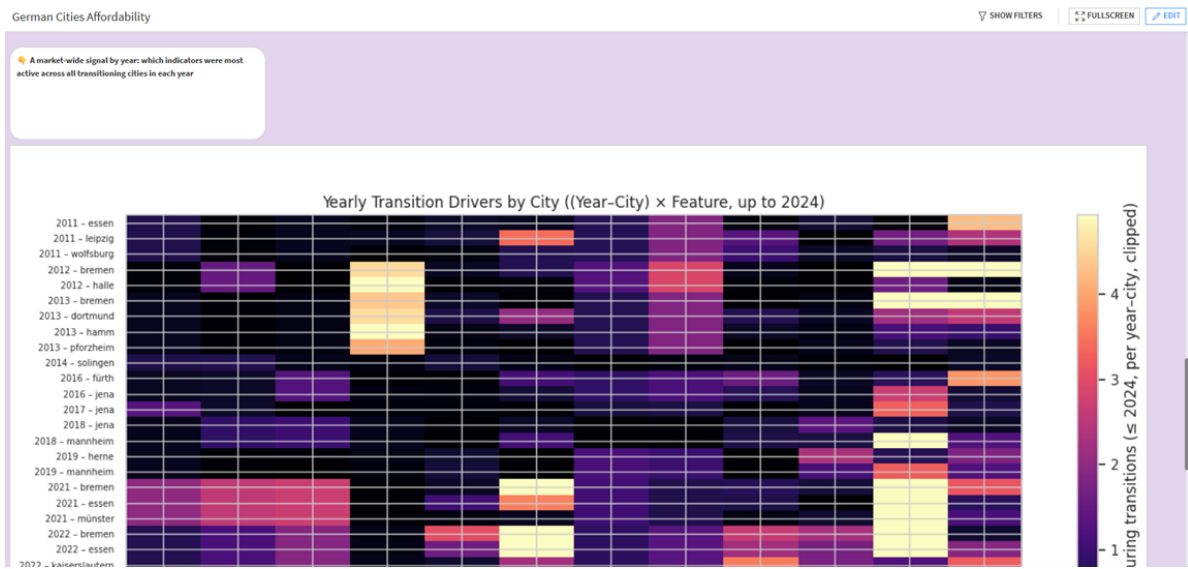


Figure A.17 – Transition drivers by city: key indicators associated with transitions (aggregate view).



**Figure A.18** – Yearly transition drivers: year-to-year feature dynamics supporting interpretation.

German Cities Affordability

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### Affordability Dashboard – Investor FAQ

This FAQ explains what the dashboard measures, how to read the results, and how the analysis supports investment decision-making across German cities over time.

- ▶ 1) What does “affordability” mean in this dashboard?
- ▶ 2) What is the key output? How should I read the clusters?
- ▶ 3) What does “transition” mean, and why is it important?
- ▶ 4) How does the dashboard explain “why” a city is in a cluster?
- ▶ 5) What is the end-to-end story from data to conclusion?
- ▶ 6) How can investors use this dashboard in practice?

Disclaimer: The dashboard provides a systematic, data-driven market typology and transition monitoring. It is intended as decision support and should be interpreted alongside asset-specific underwriting, local regulatory context, and qualitative market intelligence.

**Figure A.19** – Affordability dashboard: investor FAQ (methodology and interpretation guidance).

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